Answer Key

Other (142 Questions)

Q1. (A)	Q2. (B)	Q3. (A)	Q4. (C)	Q5. (D)
Q6. (D)	Q7. (C)	Q8. (D)	Q9. (D)	Q10. (B)
Q11. (C)	Q12. (B)	Q13. (B)	Q14. (B)	Q15. (B)
Q16. (A)	Q17. (C)	Q18. (D)	Q19. (B)	Q20. (B)
Q21. (D)	Q22. (C)	Q23. (A)	Q24. (A)	Q25. (A)
Q26. (D)	Q27. (A)	Q28. (B)	Q29. (D)	Q30. (C)
Q31. (D)	Q32. (B)	Q33. (B)	Q34. (A)	Q35. (A)
Q36. (A)	Q37. (C)	Q38. (C)	Q39. (C)	Q40. (D)
Q41. (A)	Q42. (D)	Q43. (D)	Q44. (D)	Q45. (C)
Q46. (D)	Q47. (B)	Q48. (D)	Q49. (A)	Q50. (B)
Q51. (C)	Q52. (A)	Q53. (D)	Q54. (A)	Q55. (A)
Q56. (C)	Q57. (B)	Q58. (A)	Q59. (B)	Q60. (D)
Q61. (A)	Q62. (B)	Q63. (C)	Q64. (D)	Q65. (A)
Q66. (B)	Q67. (D)	Q68. (C)	Q69. (A)	Q70. (C)
Q71. (A)	Q72. (C)	Q73. (D)	Q74. (B)	Q75. (A)
Q76. (C)	Q77. (A)	Q78. (A)	Q79. (A)	Q80. (C)
Q81. (B)	Q82. (D)	Q83. (D)	Q84. (D)	Q85. (B)
Q86. (A)	Q87. (C)	Q88. (D)	Q89. (A)	Q90. (C)
Q91. (C)	Q92. (B)	Q93. (D)	Q94. (B)	Q95. (D)
Q96. (A)	Q97. (A)	Q98. (D)	Q99. (D)	Q100.(A)
Q101.(C)	Q102.(A)	Q103.(D)	Q104.(C)	Q105.(C)

Q106.(C)	**Q107.(D)**	**Q108.(B)**	**Q109.(B)**	**Q110.(C)**
Q111.(B)	**Q112.(C)**	**Q113.(D)**	**Q114.(B)**	**Q115.(D)**
Q116.(C)	**Q117.(A)**	**Q118.(A)**	**Q119.(B)**	**Q120.(C)**
Q121.(A)	**Q122.(B)**	**Q123.(A)**	**Q124.(A)**	**Q125.(A)**
Q126.(C)	**Q127.(D)**	**Q128.(B)**	**Q129.(B)**	**Q130.(C)**
Q131.(A)	**Q132.(B)**	**Q133.(A)**	**Q134.(C)**	**Q135.(C)**
Q136.(B)	**Q137.(B)**	**Q138.(B)**	**Q139.(A)**	**Q140.(B)**
Q141.(B)	**Q142.(C)**			

Solutions

Q1. Solution

Correct Answer: (A)

$eV_0 = h(v - v_0) = hv - hv_0$

$$V_0 = \left(\frac{h}{e}\right)v - \left(\frac{h}{e}\right)v_0 \qquad \ldots(i)$$

From Eq. (i), it follows that the slope of the graph is $\frac{h}{e}$. Therefore,

$$h = e \times \text{ slope } = 1.6 \times 10^{-19} \times \frac{165 - 0}{(8-4) \times 10^{15}}$$

$$= \frac{1.6 \times 10^{-19} \times 16.5}{4 \times 10^{15}} = 6.6 \times 10^{-34} \text{ J} - \text{s}.$$

Q2. Solution

Correct Answer: (B)

Given, first resonance column at depth, $l_1 = 24$ cm

Second resonance column at depth, $l_2 = 78$ cm

Third resonance column at depths, $l_3 = ?$

As

$$l_1 + x = \frac{\lambda}{4} = 24 \qquad \ldots(i)$$

$$l_2 + x = \frac{3\lambda}{4} = 78 \qquad \ldots(ii)$$

$$l_3 + x = \frac{5\lambda}{4} \qquad \ldots(iii)$$

From Eqs. (i) and (ii),

$x = \frac{l_2 - 3l_1}{2} = \frac{78 - 3(24)}{2} = 3 \qquad \ldots(iv)$

From Eqs. (iii) and (i), $\frac{l_3 + x}{l_1 + x} = 5$

$l_3 = 5l_1 + 5x - x = 5l_1 + 4x$

Substituting l_1 and x value, we get

$l_3 = 5(24) + 4(3) \Rightarrow l_3 = 132$ cm.

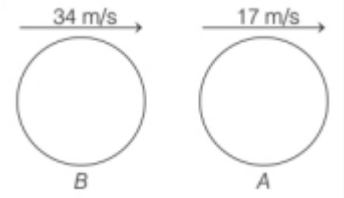
Q3. Solution

Correct Answer: (A)

$\therefore$ Given, velocity of submarine $A, v_A = 17 \text{ m/s}$

Velocity of submarine $B, v_B = 34 \text{ m/s}$

Signal sent by submarine B is detected by submarine A can be shown as



Freqency of the signal, $f_0 = 600 \text{ Hz}$

Speed of sound in water $v_s = 1500 \text{ ms}^{-1}$

Frequency received by submarine A is

$f_1 = \left(\frac{v_S - v_A}{v_S - v_B}\right) f_0 = \left[\frac{1500-17}{1500-34}\right] \times 600 \text{ Hz}$

$f_1 = \frac{1483}{1466} \times 600$

Frequency received by submarine B is

$f_2 = \left(\frac{v_S + v_B}{v_s + v_A}\right) f_1$

Substituting given values and f_1 value from Eq. (i), we get

$$f_2 = \left(\frac{1500+34}{1500+17}\right) \times \left[\frac{1483}{1466} \times 600\right]$$

$f_2 = 1.0112 \times 1.0115 \times 600$

$f_2 = 613.7 \text{ Hz}$

Q4. Solution

Correct Answer: (C)

The velocity of transverse waves is given by $v = \sqrt{\frac{T}{m}}$ where, $T =$ Tension and $m =$ mass per unit length of the wire. If r is the radius of the wire and ρ its density then,

$m = \pi r^2 \rho$

$\therefore \quad v = \sqrt{\frac{T}{m}} = \sqrt{\frac{T}{\pi r^2 \rho}}$

$\therefore \quad v_A = \frac{\sqrt{T_A}}{r_A \sqrt{\pi \rho}}$

and $\quad v_B = \frac{\sqrt{T_B}}{r_B \sqrt{\pi \rho}}$

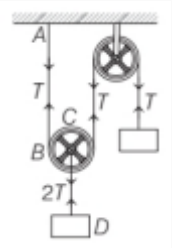
Now, $\quad \frac{v_A}{v_B} = \sqrt{\frac{T_A}{T_B}} \cdot \frac{r_B}{r_A}$

$\because \quad r_A = 2r_B$ and $T_A = \frac{1}{2} T_B$

Hence, $\quad \frac{v_A}{v_B} = \frac{1}{2\sqrt{2}}$

Q5. Solution

Correct Answer: (D)



Let the speed of wave in string be v and tension in string be T.

$$v \propto \sqrt{T}$$

$$\frac{v_{AB}}{v_{CD}} = \sqrt{\frac{T}{2T}} = \frac{1}{\sqrt{2}}$$

Q6. Solution

Correct Answer: (D)

Acceleration due to gravity at the surface of earth,

$g = \frac{GM}{R^2} = \frac{4}{3}\pi\rho GR$

$\left[\because M = \frac{4}{3}\pi R^3 \cdot \rho\right]$

Where, $\rho =$ density of earth.

Acceleration due to gravity at depth d from the surface of earth,

$g' = \frac{4}{3}\pi\rho G(R-d)$

From Eqs. (i) and (ii), we get

$g' = g\left[1 - \frac{d}{R}\right]$

Q7. Solution

Correct Answer: (C)

Given, $I = 6 + 10\sin\omega t$

$$I_{\text{eft}} = \frac{\left[\int_0^T I^2 dt\right]^{1/2}}{\int_0^T dt} = \left[\frac{1}{T}\int_0^T (6+10\sin\omega t)^2 dt\right]^{1/2}$$

$$= \left[\frac{1}{T}\int_0^T \left(36 + 120\sin\omega t + 100\sin^2\omega t\right) dt\right]^{1/2}$$

But as, $\frac{1}{T}\int_0^T \sin\omega t dt = 0$ and $\frac{1}{T}\int_0^T \sin^2\omega t = \frac{1}{2}$

$\Rightarrow \quad I_{\text{eff}} = \left[36 + \frac{1}{2}\times 100\right]^{1/2} = 9.27$

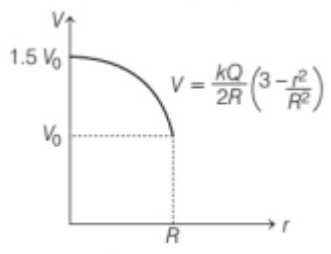
Thus, $\quad I_{\text{eff}} = 9.27$ A.

Q8. Solution

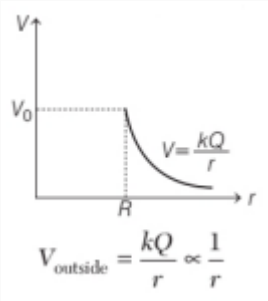
Correct Answer: (D)

Electric potential at centre $R = 0$

$V_{\text{centre}} = 1.5 V_{\text{surface}}$



where,



$$V_0 = \frac{kQ}{R}$$

$$V_{\text{inside}} = \frac{kQ}{2R}\left(3 - \frac{r^2}{R^2}\right)$$

Q9. Solution

Correct Answer: (D)

For an insulator, $E_g > 3\text{eV}$, that is why electron transition from valence band to conduction band is not possible. For semiconductor E_g is 0.2eV to 0.3eV.

Q10. Solution

Correct Answer: (B)

Work done by the gun = total kinetic energy of the bullets

$$= n\frac{1}{2}mv^2 = 300 \times \frac{1}{2} \times 10 \times 10^{-3} \times (600)^2$$

$$= 150 \times 10 \times 10^{-3} \times 600 \times 600$$

$$\text{Power of gun} = \frac{\text{work done}}{\text{Time taken}}$$

$$= \frac{150 \times 10 \times 10^{-3} \times 600 \times 600}{60 \text{ s}}$$

$$= 9 \text{ kW} \quad (\because 1 \text{ minute } = 60 \text{ seconds })$$

Q11. Solution

Correct Answer: (C)

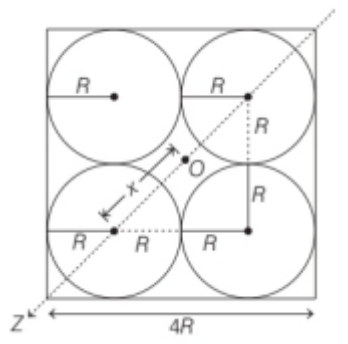
Area mass density, $\sigma = \frac{M}{16R^2}$

$\left(\because \text{ Area } = 4R \times 4R = 16R^2\right)$

Mass of each hole $m_1 = \sigma\pi R^2 = \frac{M}{16R^2}\pi R^2 = \frac{\pi M}{16}$

Distance between centre of plate and centre of hole

$x = \frac{\sqrt{(2R)^2+(2R)^2}}{2} = \frac{2\sqrt{2}R}{2}$.



Moment of inertia of one hole at about Z-axis

$I_1 = \frac{1}{2}m_1R^2 + m_1x^2 = \frac{5\pi}{32}MR^2$

Moment of inertia of whole plate about Z-axis,

$I = \frac{M(4R)^2}{6} = \frac{8}{3}MR^2$

Required moment of Inertia

$I_0 = I - 4I_1, = \left[\frac{8}{3} - 4\left(\frac{5\pi}{32}\right)\right]MR^2 = \left[\frac{8}{3} - \frac{5\pi}{8}\right]MR^2$

Given, $R = 5$ cm and $M = 1$ kg

So, $I_0 = \left[\frac{8}{3} - \frac{5\pi}{8}\right]1 \times 25 \times 10^{-4} = 0.0017 \text{ kg} - \text{m}^2$ ~

Q12. Solution

Correct Answer: (B)

According to the law of conservation of linear momentum at the highest point,

$mv\cos\theta = \frac{m}{2}(v\cos\theta) + \frac{m}{2}v'$

$\Rightarrow \quad mv\cos\theta - \frac{m}{2}v\cos\theta = \frac{m}{2}v'$

$\Rightarrow \quad v' = 2\left(v\cos\theta - \frac{v\cos\theta}{2}\right) = v\cos\theta$,

Q13. Solution

Correct Answer: (B)

$I_{\text{initial}} = 0$ for $V_i \leq +1$ V

This diode conduct only beyond $V_i = +1$ V

$I_{\text{final}} = \frac{5}{250} = 0.02$ A

So, change in $I = 0.02$ A $= 20$ mA ~

Q14. Solution

Correct Answer: (B)

The de-Broglie wavelength, $\lambda = \frac{h}{mv}$

Here, $\lambda_e = \frac{h}{m_e \frac{c}{3}}$ and $\lambda_p = \frac{h}{m_p c}$

Given, $\lambda_e = \lambda_p$

So, $\frac{h}{m_e \frac{c}{3}} = \frac{h}{m_p c}$

$\Rightarrow \frac{m_e}{m_p} = 3$

Ratio of KE $= \frac{K_e}{K_p} = \frac{\frac{1}{2} m_e v_e^2}{\frac{1}{2} m_p v_p^2} = \frac{1}{3}$

$\left(\because \frac{m_e}{m_p} = 3, v_e = \frac{c}{3} \text{ and } v_p = c\right)$,

Q15. Solution

Correct Answer: (B)

Given, $R_1 = 8\Omega$

$R_2 = 2\Omega$

$L = 5\text{mH}$

$C = 50\mu\text{F}$

and $V_0 = 20\cos(2000t)$

Impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2}$

As, here, $X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10\Omega$

Similarly, $X_L = \frac{1}{\omega C} = \frac{1}{2000 \times 50 \times 10^{-6}} = 10\Omega \because \quad X_L = X_C$, Hence, $i_{\max} = \frac{V_0}{Z} = \frac{20}{(8+2)} = 2$ A

Hence, $i_{\text{rms}} = \frac{i_{\max}}{\sqrt{2}} = \frac{2}{\sqrt{2}} = 1.41$ A and

$V = R_2 i_{\text{rms}} = 1.41 \times 2$

$= 282$ V .

Q16. Solution

Correct Answer: (A)

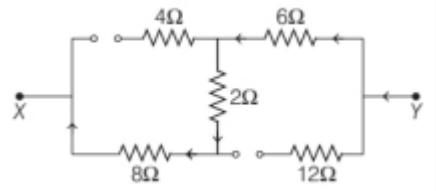
As we know that, $\frac{E}{B} = c$

so, $B = \frac{E}{c} = \frac{8}{3 \times 10^8} = 2.66 \times 10^{-8}$ ^

Q17. Solution

Correct Answer: (C)

According to the given figure X is at lower potential w.r.t Y. Hence, both diodes are in reverse biasing, so, equivalent circuit can be redrawn as follows



Equivalent resistance between X and Y

$R_{\text{eq}} = 8 + 2 + 6 = 16\Omega$ ^

Q18. Solution

Correct Answer: (D)

Let the insulated container has n moles of the monoatomic gas. The loss of kinetic energy of the gas,

$\Delta E_K = \frac{1}{2}(mn)v^2$

If the change in temperature of the gas is ΔT, then heat gained by the gas is,

$\Delta Q = \frac{3}{2}nR\Delta T$

Now, according to question,

$$\Delta Q = \Delta E_K$$

$$\frac{3}{2}nR\Delta T = \frac{1}{2}mnv^2 \text{ ^}$$

$$\Rightarrow \quad \Delta T = \frac{mv^2}{3R}$$

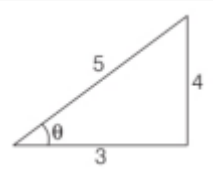
Q19. Solution

Correct Answer: (B)

Given, $\mathbf{v} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$

$\therefore \quad v = |\mathrm{v}| = \sqrt{3^2 + 4^2} \Rightarrow v = 5 \text{ m/s}$

From figure,



$\sin\theta = \frac{4}{5}$ and $\cos\theta = \frac{3}{5}$

$\because \quad R = \frac{v^2 \sin 2\theta}{g}$

$= \frac{v^2 \cdot 2\sin\theta \cdot \cos\theta}{g}$

$= \frac{5\times5\times2\times\frac{4}{5}\times\frac{3}{5}}{10} = 2.4 \text{ m !}$

Q20. Solution

Correct Answer: (B)

Given, $V_{CC} = 8$ V
Voltage drop, $V_0 = \Delta I_C R_L = 0.6$ V
Load Resistor, $R_L = 500\Omega$
Gain factor, $\alpha = 0.96$
$\because \quad \beta = \frac{\alpha}{1-\alpha} = \frac{0.96}{1-0.96} = 24$
Now, $\quad V_0 = \Delta I_C R_L$
$0.6 = \Delta I_C \times 500$
$\Delta I_C = \frac{0.6}{500} = 1.2 \times 10^{-3}$ A
Now,
$\beta = \frac{\Delta I_C}{\Delta I_B}$
$\therefore$ Base current $\Delta I_B = \frac{\Delta I_C}{\beta} = \frac{1.2 \times 10^{-3}}{24} = 5 \times 10^{-5}$
$= 50 \times 10^{-6}$
$\Delta I_B = 50 \mu$A ,

Q21. Solution

Correct Answer: (D)

Given, mass of solid sphere $= 80$ kg radius of solid sphere, $R_s = 15$ m
radius of circular disc, $R_c = 20$ m and time $= 1$ hour $= 60$ minute $= 60 \times 60$sec
$\therefore$ Moment of inertia of solid sphere, $I_s = \frac{2}{5} M R^2$

$$= \frac{2}{5} \times 80 \times (15)^2$$
$$= 7200 \text{ kg} - \text{m}^2$$

Similarly, moment of inertia of the disc,

$$I_c = \frac{1}{2} M R_c^2$$
$$= \frac{1}{2} \times 80 \times (20)^2$$
$$= 16000 \text{ kg} - \text{m}^2$$

$$\begin{aligned} \text{Rate of change of moment of Inertia } &= \frac{I_c - I_s}{t} \\ &= \frac{16000 - 7200}{60 \times 60} \\ &= \frac{22}{9} \text{ kg} - \text{m}^2 \text{ s}^{-1} \end{aligned}$$

Q22. Solution

Correct Answer: (C)

For permanent magnet we prefer a material with high retentivity and high coercivity. For electromagnet, we prefer high saturated magnetism, low coercivity and least possible area of hystersis loop. Therefore, X is suitable for making permanent magnet and Y for making electromagnet.

Q23. Solution

Correct Answer: (A)

From radioactive decay law,

$-\frac{dN}{dt} \propto N$ or $-\frac{dN}{dt} = \lambda N$

Thus, $A = -\frac{dN}{dt}$ or $A = \lambda N$

$\Rightarrow \quad A = \lambda N_0 e^{-\lambda t}$

Where, $A_0 = \lambda N_0$ is the activity of the radioactive material at time, $t = 0$

At time, $t_1 \quad A_1 = A_0 e^{-\lambda t_1}$

At time, $t_2 \quad A_2 = A_0 e^{-\lambda t_2}$

On dividing Eq. (ii) by Eq. (iii), we have

$$\frac{A_1}{A_2} = \frac{e^{-\lambda t_1}}{e^{-\lambda t_2}} = e^{-\lambda(t_1 - t_2)}$$

$\Rightarrow \quad A_1 = A_2 e^{-\lambda(t_1 - t_2)}$

Q24. Solution

Correct Answer: (A)

As, we know, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$$\Rightarrow \quad \frac{1.5}{v} - \frac{1}{(-20)} = \frac{1.5 - 1}{+40}$$

$v = -40$ cm

Negative sign shows that image is obtained on the same side of object i.e., towards left.

Q25. Solution

Correct Answer: (A)

In the curve, $V \propto \frac{1}{T}$

$\Rightarrow V = m\frac{1}{T}$ where, $m =$ slope of curve.

$\Rightarrow VT =$ constant

$\therefore \quad TdV + VdT = 0$

$\Rightarrow \quad dV = -\frac{V}{T}dT$

As, $Q = \Delta U + W$

$CdT = C_V dT + pdV$

$CdT = C_V dT - \frac{p}{T}VdT$

$\Rightarrow \quad C = C_V - \frac{pV}{T}$

$= C_V - R \quad \left(\because \frac{pV}{T} = R\right)$

For diatomic gas, $C_V = \frac{5}{2}R$

$\therefore \quad C = C_V - R = \frac{5}{2}R - R = \frac{3R}{2}$

Q26. Solution

Correct Answer: (D)

Liquid volume coming out of the tube per second

$$V = \frac{p\pi r^4}{8\eta l}$$

$\Rightarrow$

$$\Rightarrow \frac{V_2}{V_1} = \left[\frac{r_2}{r_1}\right]^4 \Rightarrow V_2 = V_1\left[\frac{125}{100}\right]^4 = V_1\left(\frac{5}{4}\right)^4$$

$\Rightarrow$

So, now $\frac{\Delta V}{V} = \frac{V_2 - V_1}{V_1} = \frac{(244)V_1 - V_1}{V_1} = 144\%$

Q27. Solution

Correct Answer: (A)

When S_2 open.

Assume resistance of $XY = R$.

Resistance of wire per unit length, $x = \frac{R}{L} = R\Omega\mathrm{m}^{-1}$

$\because \quad I = E_0/R$

Now, the potential drop across 50 cm length is 6 V, So

$\frac{E_0}{R} \times R \times \frac{50}{100} = 6$

$\Rightarrow \quad E_0 = 12$ V

When S_2 closed, potential drop across 0.416 cm length,

$V_1 = \frac{E_0}{R} \times R \times 0.416 = 12 \times 0.416 \approx 5$ V

Hence, $E - Ir = 5$ V

$\Rightarrow \quad 6 - Ir = 5$

$\because \quad I = \frac{5}{10}$

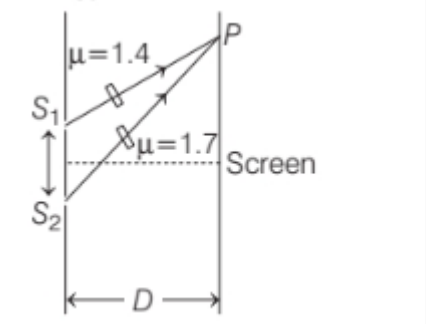
$\therefore \quad 6 - 5 = \frac{5}{10}r$

$\Rightarrow \quad r = 2\Omega$

Q28. Solution

Correct Answer: (B)

As, shift $\Delta x = \frac{\beta}{\lambda}(\mu - 1)t$



Shift due to one plate $\Delta x_1 = \frac{\beta}{\lambda}(\mu_1 - 1)t$

Shift due to another path $\Delta x_2 = \frac{\beta}{\lambda}(\mu_2 - 1)t$

Net shift $\Delta x = \Delta x_2 - \Delta x_1 = \frac{\beta}{\lambda}(\mu_2 - \mu_1)t$

Also, it is given that $\Delta x = 6\beta$

Hence, $6\beta = \dfrac{\beta}{\lambda}(\mu_1 - \mu_2)t$

$$\Rightarrow \quad t = \frac{6\lambda}{(\mu_1 - \mu_2)} = \frac{6 \times 4000}{(1.7 - 1.4)} - 80000\mathring{A}$$

$$\Rightarrow \quad t = 8 \times 10^{-6} = 8\mu\text{m}$$

Q29. Solution

Correct Answer: (D)

Force on a moving charged particle in uniform magnetic field

$F = Bqv \sin\theta$

Since, charge particle moves along the axis of circular current carrying loop, therefore, $\theta = 0^\circ$ or 180°

When $\theta = 0^\circ; F = Bqv \sin 0^\circ$

$F = 0$

When $\theta = 180^\circ, F = Bqu \sin 180^\circ$

$F = 0$

Q30. Solution

Correct Answer: (C)

- The condition of achromatism is $W_1P_1 + W_2P_2 = 0$

$\Rightarrow \quad W_1P_1 \quad = -W_2P_2$

$\Rightarrow \qquad \frac{W_1}{W_2} \quad = -\frac{P_2}{P_1}$

Now, $\quad P_1 + P_2 = 4\mathrm{D}$

but, Power of converging lens,

$P_1 = 5\mathrm{D}$

$\because$ Power of diverging lens

$P_2 = 4\mathrm{D} - P_1$

$\quad = 4\mathrm{D} - 5\mathrm{D} = -\mathrm{D}$

[From ii]

$\therefore$ From Eq. (i), we have

$\frac{W_1}{W_2} = -\frac{P_2}{P_1} = \frac{-(-D)}{5D} = \frac{1}{5} \Rightarrow \frac{W_1}{W_2} = \frac{1}{5}$

Q31. Solution

Correct Answer: (D)

Protium (P), deuterium (D) and tritium (T) are the three isotopes of hydrogen. These isotopes follow the order in different contexts as shown below :

$T_2 > D_2 > P_2$ [Order of boiling point (BP)]

$T_2 > D_2 > P_2$ [Order of bond energy (BE)]

$T_2 = D_2 = P_2$ [Order of bond length (BL)]

$T_2 < D_2 < P_2$ [Order of reactivity with Cl_2]

Hence, all the given order are correct.

Q32. Solution

Correct Answer: (B)

The amino acid which gives yellow colour with ninhydrin in paper chromatography is proline.

Ninhydrin + Proline $\longrightarrow$ Yellow-orange product

Q33. Solution

Correct Answer: (B)

In case of liquids, cohesive forces decrease and hence, their viscosity decreases while increasing the temperature. However, in case of gases, with the increase in temperature kinetic energy of gaseous molecules increases and the molecules become more mobile.

Q34. Solution

Correct Answer: (A)

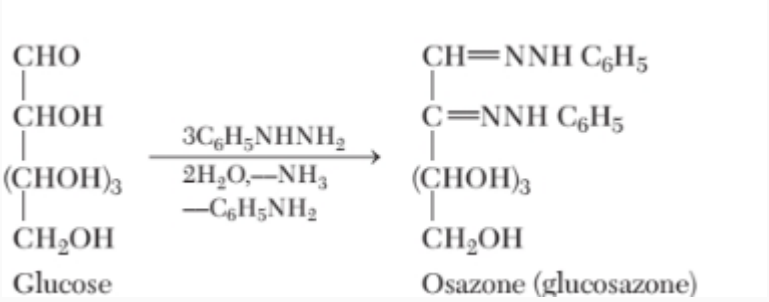
The thermal stability of carbonates increases down the group. So, the order of thermal stability of carbonates of alkaline earth metals is

$BaCO_3 > CaCO_3 > MgCO_3 > BeCO_3$.

Therefore, $BaCO_3$ is thermodynamically most stable and $BeCO_3$ is least stable.

Q35. Solution

Correct Answer: (A)

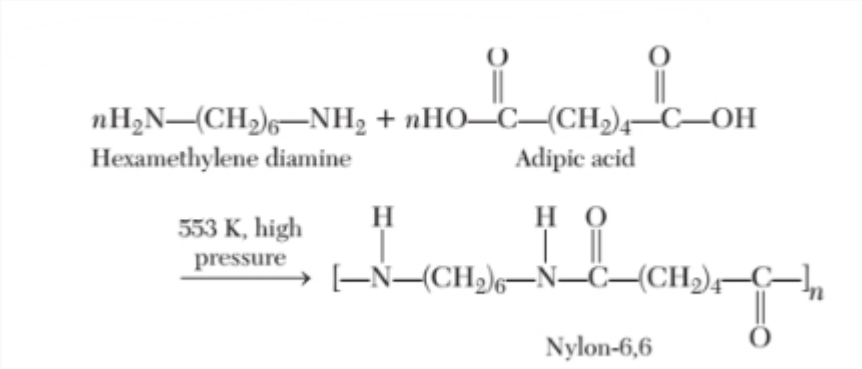


Hence, three molecules of phenyl hydrazine $(C_6H_5NHNH_2)$ is used to yield osazone.

Q36. Solution

Correct Answer: (A)

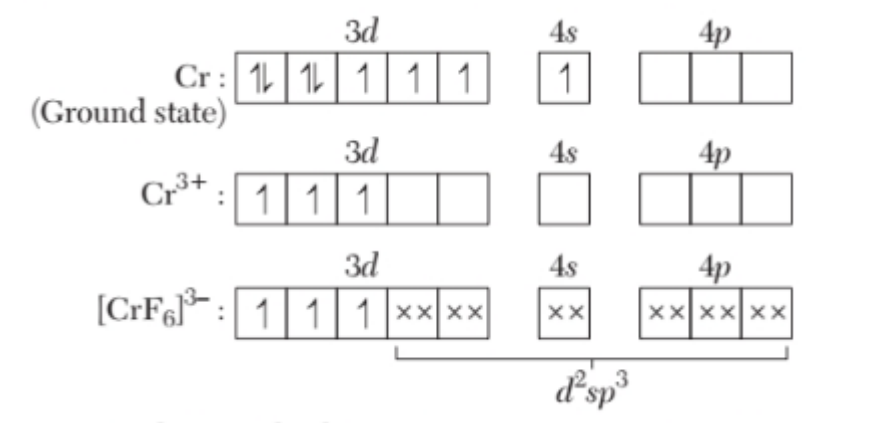
The monomer units of nylon- 6,6 are hexamethylene diamine and adipic acid.

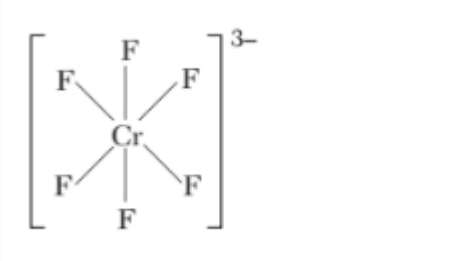


Q37. Solution

Correct Answer: (C)

The electronic configuration of Cr is $[\mathrm{Ar}]3d^5 4s^1$.





$[\mathrm{CrF_6}]^{3-}$ has d^2sp^3-hybridisation and octahedral geometry.

Q38. Solution

Correct Answer: (C)

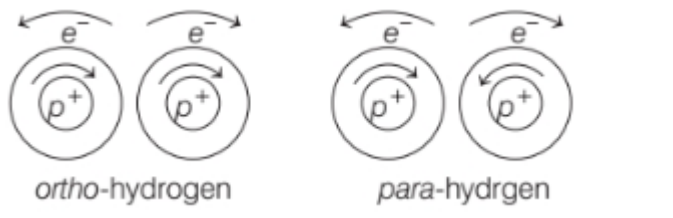
Species	Electrons	Bond order	Unpaired electron	
O—F	17	1.5	1	Paramagnetic
F_2	18	1.0	0	Diamagnetic

This is one unpaired electron in $(\mathrm{O}-\mathrm{F})$, hence paramagnetic, thus (a) is correct.
As bond order of $\mathrm{O}-\mathrm{F} > \mathrm{F}-\mathrm{F}$ thus, bond energy of $\mathrm{O}-\mathrm{F} > \mathrm{F}-\mathrm{F}$.
Hence, (a) and (b) both are correct.

Q39. Solution

Correct Answer: (C)

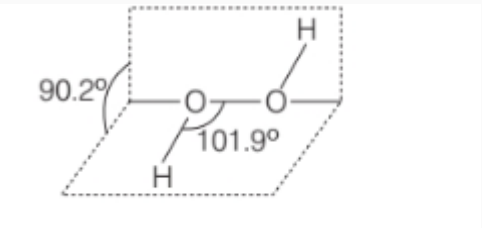
Ortho and para hydrogen similar to each other in their chemical properties but they have different physical properties like boiling point, thermal conductivity, due to their difference in overall spins.



Q40. Solution

Correct Answer: (D)

Hydrogen peroxide (H_2O_2) is a non-planar molecule with twisted symmetrical structure. It has non-linear structure with an open book structure.



Q41. Solution

Correct Answer: (A)

The correct match is

$A \rightarrow (4); B \rightarrow (3); C \rightarrow (2); D \rightarrow (1)$

A. DDT $\rightarrow$ Pesticides

B. $NaClO_3 \rightarrow$ Herbicides

C. $Cl_2 \rightarrow$ Disinfectant

D. PAN $\rightarrow$ Photochemical smog

Q42. Solution

Correct Answer: (D)

		Bond order	
I.	O_2^{2-}	1	Weaker
	O_2	1	
II.	N_2	3	Stronger
	N_2^+	2.5	
III.	NO^+	3	Stronger
	NO^-	2	

Larger the bond order, stronger the bond formed.

Q43. Solution

Correct Answer: (D)

$$[Co(NH_3)_5SO_4]Br \rightleftharpoons \underset{\substack{\uparrow \\ \text{Yellow ppt.} \\ \text{with } AgNO_3}}{[Co(NH_3)_5SO_4]^+} + Br^-$$

$$[Co(NH_3)_5Br]SO_4 \rightleftharpoons [Co(NH_3)_5Br]^{2+} + \underset{\substack{\uparrow \\ \text{White ppt. with } BaCl_2}}{SO_4^{2-}}$$

Q44. Solution

Correct Answer: (D)

$MS_2 = M + 32 \times 2 = M + 64$

% of sulphur $= \left(\frac{64}{M+64}\right) \times 100 = 40.06$

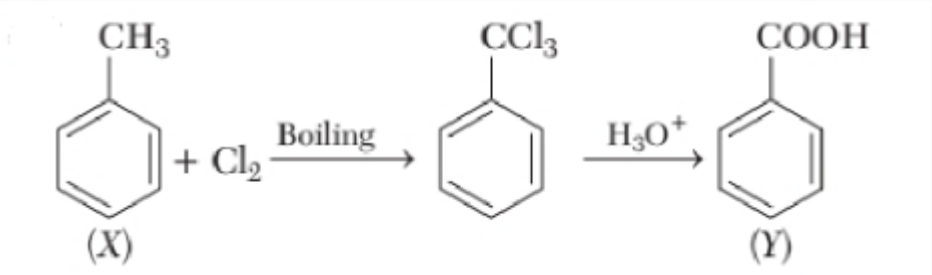
$M + 64 = \frac{6400}{40.06}$

$M + 64 = 160$

$M = 160 - 64 = 96$ u

Q45. Solution

Correct Answer: (C)



Q46. Solution

Correct Answer: (D)

Inert pair effect is more pronounced in heavier members like Pb.
Hence, Pb (IV) compounds act as strong oxidising agents and are reduced to more stable Pb (II) compounds.

Q47. Solution

Correct Answer: (B)

Anti-fluorite structure refers to an anion array with tetrahedral cations and the compound having A_2B formula have anti-fluorite structure. So, $\mathrm{Na_2O}$ has an anti-fluorite structure in which $\mathrm{Na^+}$ions occupy all the tetrahedral voids and $\mathrm{O^{2-}}$occupy half of the cubic holes.

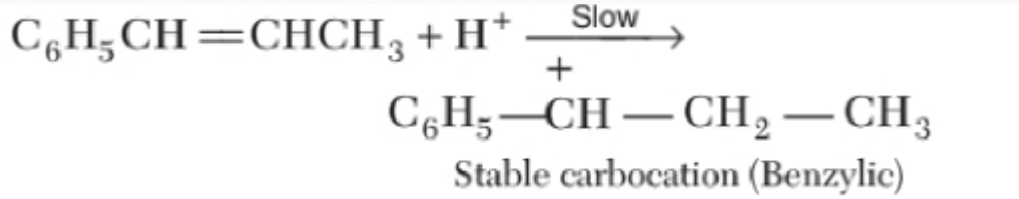
Q48. Solution

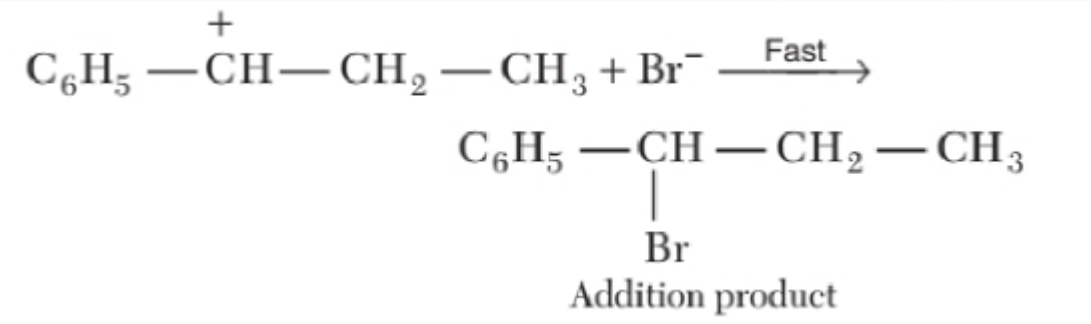
Correct Answer: (D)

Sodium formate is present at the equivalence point. It is the salt of weak acid $+$ strong base. So, final solution will be basic in nature.
As we know
$\mathrm{pH} = 7 + \frac{\mathrm{p}K_a}{2} + \frac{\log C}{2}$
where, C is concentration of salt.
Total volume of solution $= 100 + 100 = 200$ mL
Concentration of salt $(C) = 2 \times \frac{100}{200} = 1\mathrm{M}$
$\mathrm{pH} = 7 + \frac{3.74}{2} + \frac{\log[1]}{2}$
$\mathrm{pH} = 7 + 1.87 + 0$
$\mathrm{pH} = 8.87$

Q49. Solution

Correct Answer: (A)



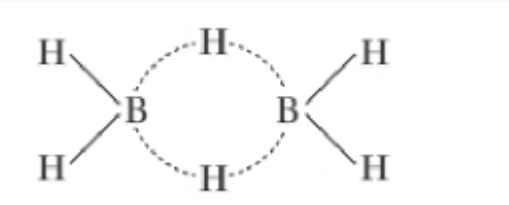


Electrophile addition reaction takes place via more stable carbocation.

Q50. Solution

Correct Answer: (B)

The structure of diborane is



In this structure, there are two $3C-2e^-$bonds. 3 atoms, $B-H-B$ share two electrons and form an angular geometry, leading bent bond, also known as banana bond.

Q51. Solution

Correct Answer: (C)

$\Delta S = S(XY_3) - \frac{1}{2}S(X_2) - \frac{3}{2}S(Y_2)$

$= 50 - 30 - 60 = -40 \text{ J mol}^{-1} \text{ K}^{-1}$

$\Delta H = -30 \text{ kJ} = -3000 \text{ J}$

$\Delta G = \Delta H - T\Delta S$

At equilibrium,

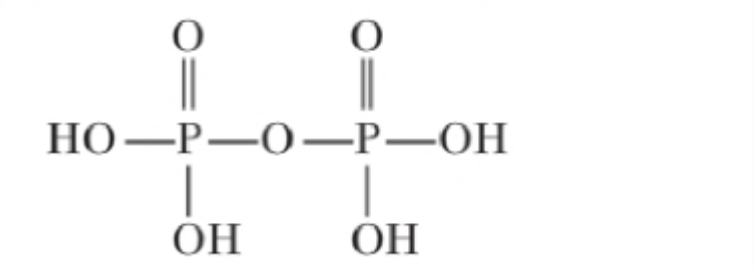
$\Delta G = 0$

$T = \frac{\Delta H}{\Delta S} = \frac{-3000}{-40} = 750 \text{ K}$

Q52. Solution

Correct Answer: (A)

The structure of pyrophosphoric acid is



There are $4P-OH$ bonds, $1P-O-P$ bond and two $P=O$ bonds.

Q53. Solution

Correct Answer: (D)

(a)

$$2Fe^{3+} + 2e^- \longrightarrow 2Fe^{2+};\ E^\circ = 0.77\ V$$

$$2I^- \longrightarrow I_2 + 2e^-;\ E^\circ = -0.54\ V$$

(sign of E° is reversed)

$$2Fe^{3+} + 2I^- \longrightarrow 2Fe^{2+} + I_2;\ E^\circ_{cell} = +0.23\ V$$

This reaction is feasible, since E°_{cell} is positive.

(b)

$$Cu \longrightarrow Cu^{2+} + 2e^-;\ E^\circ = -0.34\ V$$

(sign of E° has been reversed)

$$2Ag^+ + 2e^- \longrightarrow 2Ag;\ E^\circ = +0.80\ V$$

$$Cu + 2Ag^+ \longrightarrow Cu^{2+} + 2Ag;\ E^\circ = +0.46\ V$$

This reaction is feasible, since E°_{cell} is positive.

(c)

$$2Fe^{3+} + 2e^- \longrightarrow 2Fe^{2+};\ E^\circ = +0.77\ V$$

$$Cu \longrightarrow Cu^{2+} + 2e^-;\ E^\circ = -0.34\ V$$

(sign of E° is reversed)

$$2Fe^{3+} + Cu \longrightarrow 2Fe^{2+} + Cu^{2+};\ E^\circ = +0.43V$$

This reaction is feasible, since E°_{cell} is positive.

(d)

$$Ag \longrightarrow Ag^+ + e^-;\ E^\circ = -0.80\ V$$

(sign of E° is reversed)

$$Fe^{3+} + e^- \longrightarrow Fe^{2+};\ E^\circ = +0.77\ V$$

$$Ag + Fe^{3+} \longrightarrow Ag^+ + Fe^{2+};\ E^\circ = -0.03V$$

This reaction is not feasible, since E°_{cell} is negative.

Q54. Solution

Correct Answer: (A)

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]}$$

$$1.8 \times 10^{-5} = \frac{[NH_4^+][0.01]}{[0.02]}$$

$$\therefore \quad [NH_4^+] = 3.6 \times 10^{-5} M$$

Q55. Solution

Correct Answer: (A)

van't Hoff equation giving variation of equilibrium constant, K with temperature T is

$\log_e K = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$

Given, $\quad \log_e K = -\frac{2000}{T} + 4.0$

$\therefore \quad \Delta S^\circ / R = 4 \Rightarrow \Delta S^\circ = 4R$

Q56. Solution

Correct Answer: (C)

Free expansion, $W = 0$

Adiabatic process, $q = 0$

$\Delta U = q + W = 0$, this means that internal energy remains constant. Therefore, $\Delta T = 0$ in ideal gas there is no intermolecular attraction.

Hence, when such a gas expands under adiabatic conditions into a vacuum, no heat is absorbed or evolved, since no external work is done to separate the molecules.

Q57. Solution

Correct Answer: (B)

From the graph it is clear that, the value of ' Z ' decreases with increase of pressure. We can explain as follows on the basis of van der Waals' equation.

At high pressure, when ' p ' is large, V will be small and one cannot ignore ' b ' in comparison to V. However, the team a/V^2 may be considered negligible in comparison to ' p ' in van der Waals' equation.

$\left(p + \frac{a}{V^2}\right)(V - b) = nRT$

$p(V - b) = nRT$

$\Rightarrow \quad pV - pb = nRT$

or $\frac{pV}{nRT} = 1 + \frac{pb}{nRT}$

or $Z = 1 + \frac{pb}{nRT}$

Thus, Z is greater than 1. As pressure is increased (at constant T), the factor $\frac{pb}{nRT}$ increases. This explains why after minima in the curves, Z increase continuously with pressure. Hence, the only incorrect statement is (b).

Q58. Solution

Correct Answer: (A)

The hydrogen atom has the configuration $1s$ and $3s, 3p$ and $3d$-orbitals will have same energy with respect to $1s$-orbital.

Q59. Solution
Correct Answer: (B)

CH_4; bp $= 4$; sp^3-hybridisation
NF_3; bp $= 3$; $lp = 1$; sp^3-hybridisation
NH_4^+; bp $= 4$; $lp = 0$; sp^3-hybridisation
H_2O; bp $= 2$; $lp = 2$; sp^3-hybridisation
Hence, all have the same slope.

Q60. Solution
Correct Answer: (D)

Given,
Mass of an organic compound $= 0.20$ g
Mass of $AgBr = 0.12$ g
Molecular mass of $AgBr = 188$ g mol^{-1}
188 g of AgBr contains 80 g of bromine
$\therefore$ 0.12 g of $AgBr$ will contain $= \frac{80}{188} \times 0.12$
$= 0.05$ g of bromine
$\therefore$ Percentage of bromine $= \frac{0.05}{0.20} \times 100$
$= 25\%$

Q61. Solution
Correct Answer: (A)

Forthrightness and 'outspokenness' both have a same meaning, i.e. straight forward.

Q62. Solution
Correct Answer: (B)

'Inexorable' means 'relentless', i.e. without any particular end.

Q63. Solution
Correct Answer: (C)

'Then she bowed her face', is the correct passive form of the given sentence.

Q64. Solution
Correct Answer: (D)

The complex form of the sentence would be 'the child is spoiled when the rod is spared'.

Q65. Solution
Correct Answer: (A)

Retrograde means declining. Hence, 'progressive' is its correct antonym.

Q66. Solution
Correct Answer: (B)

'Covert' means hidden. So, 'obvious' would be its correct antonym.

Q67. Solution
Correct Answer: (D)

Disinterested and selfless concern for the well-being of other is called Altruism.

Q68. Solution
Correct Answer: (C)

Code of diplomatic etiquette and precedence is called protocol.

Q69. Solution
Correct Answer: (A)

Explanation: (B) provides the background: "In India, corporations until recently achieved success by avoiding competition, using protected and regulated domestic markets." This sets the stage for the transition from a protected market to a competitive one. (A) follows logically: "Now under a liberated economy, they are learning to compete domestically and globally." This contrasts the past (B) with the present (A). (D) explains how businesses are reacting: "Business leaders are preparing themselves to meet competitive challenges, and to avoid being swept away." This connects well after (A), showing how companies are adapting. (C) concludes by stating the inevitability: "The trend is irreversible." This wraps up the paragraph by emphasizing that competition is here to stay. Thus, the correct logical sequence is BADC (Option 1)

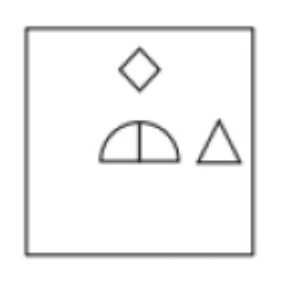
Q70. Solution
Correct Answer: (C)

CBAD is the correct sequence to form a meaningful and logical paragraph.

Q71. Solution

Correct Answer: (A)

In the given series triangle is moving $\frac{1}{2}$ place in clockwise direction along the sides and corners of square and also unshaded and shaded alternatively. The rhombus is moving one place in anti-clockwise direction along the sides of square. Hence, we will get the following figure.

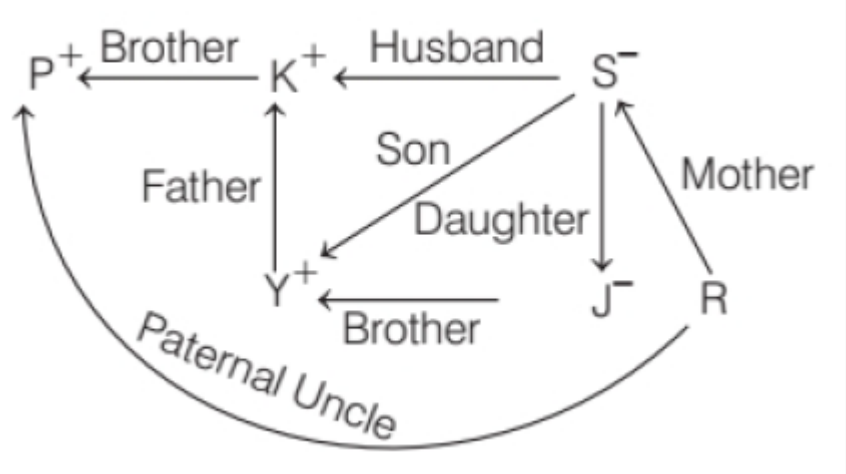


Q72. Solution

Correct Answer: (C)

Let us draw the family diagram.

$P - K \times Y - J \div S + R \Rightarrow$

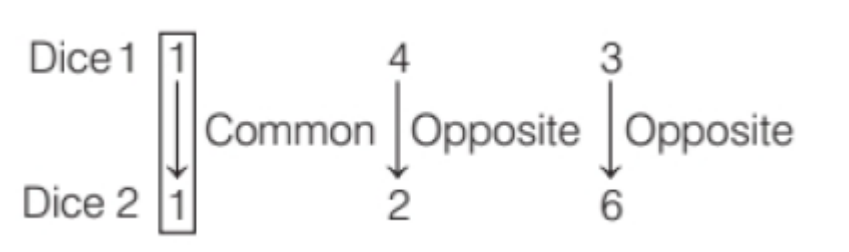


Clearly, option (c) is wrong because J is the daughter of K and S.

Q73. Solution

Correct Answer: (D)

1 number is common in first two position of dices. Now, start with 1 number, writing all the remaining numbers in clockwise direction.



Hence, 3 number will be on the face opposite to the one showing 6 .

Q74. Solution

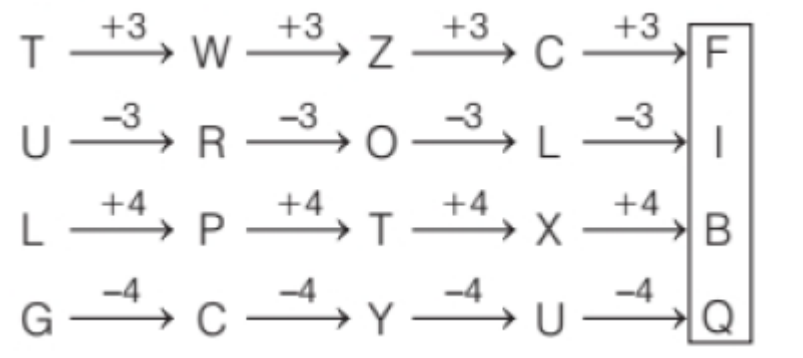
Correct Answer: (B)

The question figure is embedded in answer figure (b).

Q75. Solution

Correct Answer: (A)

The pattern of the series is as follows

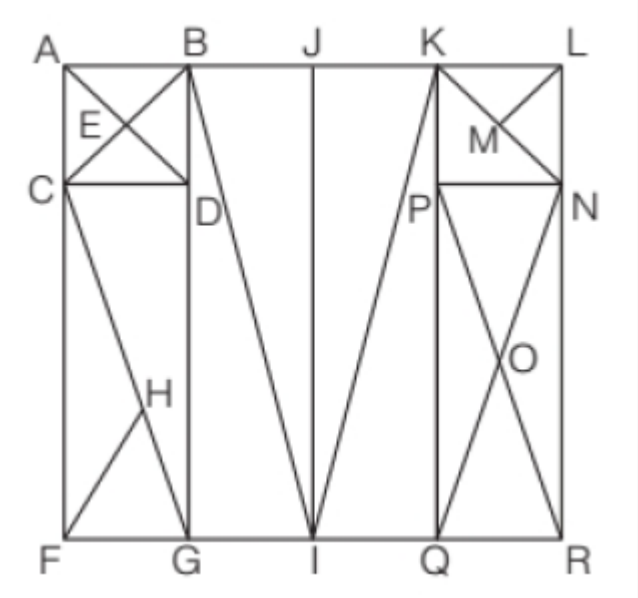


$\therefore ? =$ FIBQ

Q76. Solution

Correct Answer: (C)

The number of triangles in the figure are:



$\Delta ABE, \Delta AEC, \Delta CED, \Delta BED, \Delta ADC, \Delta CDB, \Delta ABC, \Delta ABD, \Delta PON, \Delta QOR, \Delta POQ, \Delta NOR, \Delta PNQ, \Delta PQR, \Delta NQR, \Delta NPR, \Delta CFG, \Delta CDG, \Delta CHF, \Delta HFG, \Delta KPN, \Delta KLN, \Delta KLM, \Delta LMN, \Delta BGI, \Delta BIJ, \Delta JIK, \Delta BIK, \Delta KIQ, \Delta KNQ$ and $\Delta BCG = 31$

Q77. Solution

Correct Answer: (A)

Average marks of 50 students $= 64$

Sum of marks of 50 students

$= \text{ Average } \times \text{ number of students}$

$= 64 \times 50 = 3200$

According to the question,

Correct sum of marks of 50 students

$= 3200 - 38 - 42 + 83 + 24$

$= 3200 + 27$

$= 3227$

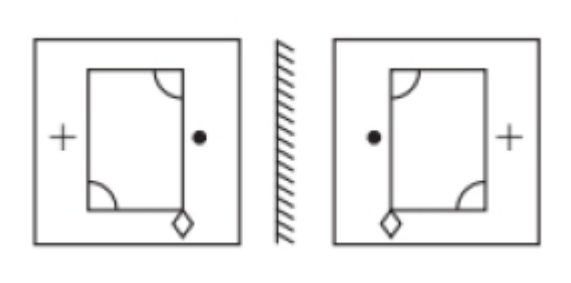
$\therefore$ Correct average marks of 50 students

$= \frac{3227}{50} = 64.54$

Q78. Solution

Correct Answer: (A)

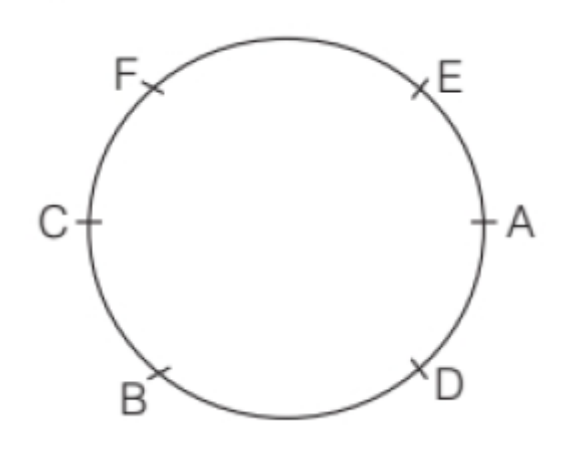
When a mirror is placed on the right side of the image,
The facing of the figure to the left and right direction changes to the opposite direction in the mirror image. There will be change in top and bottom direction. The mirror image of the given figure is



Q79. Solution

Correct Answer: (A)

According to the question,



Hence, A sits immediate left of E.

Q80. Solution

Correct Answer: (C)

According to the question,

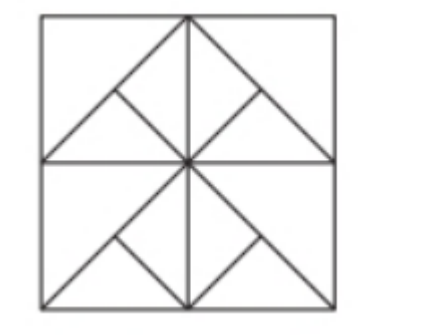
$$E > \boxed{B} > C > A > D$$

Hence, B's car was the 2nd costliest.

Q81. Solution

Correct Answer: (B)

If figure shown in option (b), is placed in the place of missing portion of the original figure, then it is completed as shown below



Q82. Solution

Correct Answer: (D)

Clearly, in the first row, the number of dots in the second figure is twice the number of dots in the first figure. Similarly, in the second row, the number of dots in the second figure must be twice the number of dots in the first figure.

So, number of dots in missing segment $= 2 \times 3 = 6$

Q83. Solution

Correct Answer: (D)

Either of the situation is possible. If Mr. Gopal was one of the member of $60^{\wedge}$ employees, then he went on strike. If he was not in group of 60%, then he did not participate in the strike. Hence, either conclusion I or II follows.

Q84. Solution

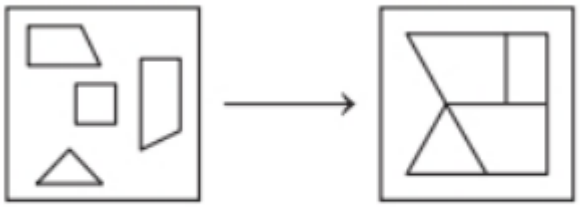
Correct Answer: (D)

The statement I is talking about in a condition with the lawyer that they marry only fair girls. But it is not talking about any condition with Shobha. So, Shobha can marry either a lawyer or anyone else.

Q85. Solution

Correct Answer: (B)

Figure given in option (b) can be formed by
joining the pieces given in question figure, as shown below.



Q86. Solution

Correct Answer: (A)

As, pressure measure in Borometer. Similarly, current measure in Ammeter.

Q87. Solution

Correct Answer: (C)

As Cat, Lion and Jaguar belongs to the same species called Cat. Similarly, Monkey, Gorilla and Chimpanzee belong to the same species of animal called Apes.

Q88. Solution

Correct Answer: (D)

As, 'Needle' is used for 'Sewing'. Similarly, 'Microscope' is used for 'Magnifying'.

Q89. Solution

Correct Answer: (A)

As, $14 : 289 \Rightarrow 14 + 3 = 17$

$\Rightarrow \quad 17^2 = 289$

And $17 : 400 \Rightarrow 17 + 3 = 20$

$\Rightarrow \quad 20^2 = 400$

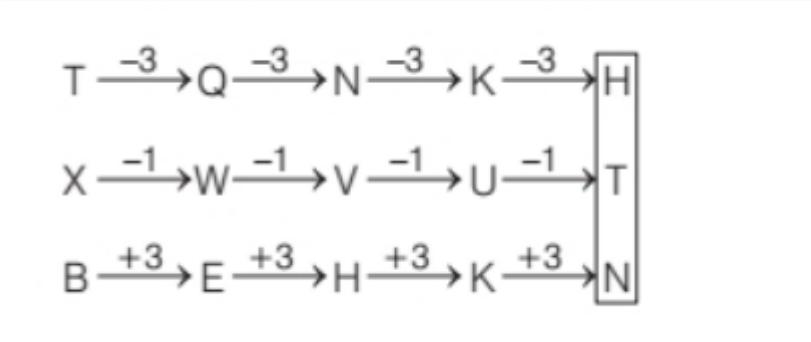
Similarly, $\quad 21 :? \Rightarrow 21 + 3 = 24$

$\Rightarrow \quad 24^2 = 576$

Q90. Solution

Correct Answer: (C)

The pattern of the series is as follows,



$? = \text{HTN}$

Q91. Solution

Correct Answer: (C)

$4x^2 + 2x - 1 = 0$

$\therefore \quad \alpha + \beta = -\frac{1}{2}, \alpha\beta = -\frac{1}{4}$

Also, $4\alpha^2 + 2\alpha - 1 = 0$ as α is a root and we have to prove that $\beta = 4\alpha^3 - 3\alpha$.

Now, $4\alpha^3 - 3\alpha = 4\alpha^2 \cdot \alpha - 3\alpha$

$= \alpha(1 - 2\alpha) - 3\alpha = -2\alpha^2 - 2\alpha$

$= -\frac{1}{2}\left[4\alpha^2 + 4\alpha\right] = -\frac{1}{2}[1 - 2\alpha + 4\alpha]$

$= -\frac{1}{2}(1 + 2\alpha) = -\frac{1}{2} - \alpha = \beta$

Now, $\alpha + \beta = -\frac{1}{2}$

[from Eq. (i)]

Hence, $\beta = 4\alpha^3 - 3\alpha$.

Q92. Solution

Correct Answer: (B)

Let $x \in A \cap B$. Then,

$x \in A$ and $x \in B$

$\Rightarrow x$ is a multiple of 4 and x is a multiple of 6

$\Rightarrow x$ is a multiple of 4 and 6 both

$\therefore x$ is a multiple of 12

Q93. Solution

Correct Answer: (D)

We have, $|w| = 2 \Rightarrow w = 2(\cos\theta + i\sin\theta)$

$\therefore \quad z = w - \frac{1}{w}$

$= 2(\cos\theta + i\sin\theta) - \frac{1}{2}(\cos\theta - i\sin\theta)$

$\Rightarrow \quad z = \frac{3}{2}\cos\theta + \frac{5}{2}i\sin\theta$

$\Rightarrow \quad |z| = \sqrt{\frac{9}{4} + \frac{25}{4}} = \frac{\sqrt{17}}{2} < 3$

Hence, option (d) is correct.

Q94. Solution

Correct Answer: (B)

$1 - \cos(1 - \cos x) = 2\sin^2\left(\frac{1-\cos x}{2}\right)$

$= 2\sin^2\left(\sin^2\frac{x}{2}\right)$

$\therefore \lim_{x\to 0} \frac{1-\cos(1-\cos x)}{x^4} = \lim_{x\to 0} \frac{2\sin^2\left(\sin^2\frac{x}{2}\right)}{x^4}$

$= \lim_{x\to 0} \frac{2\sin^2\left(\sin^2\frac{x}{2}\right)}{\left(\sin^2\frac{x}{2}\right)^2} \times \frac{\sin^4\frac{x}{2}}{\left(\frac{x}{2}\right)^4 \times 16}$

$= \frac{1}{8}$

Q95. Solution

Correct Answer: (D)

Let d be the common difference of the AP. Then,

$a_{10} = 3 \Rightarrow a_1 + 9d = 3$

$\Rightarrow 2 + 9d = 3 \Rightarrow d = \dfrac{1}{9}$

$\therefore a_4 = a_1 + 3d = 2 + \dfrac{1}{3} = \dfrac{7}{3}$

Let D be the common difference of $\frac{1}{h_1}, \frac{1}{h_2}, \ldots . -\frac{1}{h_{10}}$.

Then, $\quad h_{10} = 3$

$\Rightarrow \quad \frac{1}{h_{10}} = \frac{1}{3} \Rightarrow \frac{1}{2} + 9D = \frac{1}{3}$

$\Rightarrow \quad 9D = -\frac{1}{6} \Rightarrow D = -\frac{1}{54}$

$\therefore \quad \frac{1}{h_7} = \frac{1}{h_1} + 6D = \frac{1}{2} - \frac{1}{9} = \frac{7}{18}$

$\Rightarrow \quad h_7 = \frac{18}{7}$

$\therefore \quad a_4 h_7 = \frac{7}{3} \times \frac{18}{7} = 6$

Q96. Solution

Correct Answer: (A)

We have,

$(1 + 5\sqrt{2}x)^9 + (1 - 5\sqrt{2}x)^9$

$= 2\left\{{}^2C_0 + {}^9C_2(5\sqrt{2}x)^2 + \ldots + {}^9C_8(5\sqrt{2}x)^8\right\}$

Clearly, it has 5 terms.

Q97. Solution

Correct Answer: (A)

Other than 2, remaining five places can be filled by 1 and 3 for each place. The number of ways for five places is $2 \times 2 \times 2 \times 2 \times 2 = 2^5$. For 2 , selecting 2 places out of 7 is ${}^7C_2$. Hence, the required number of ways is ${}^7C_2 \times 2^5$.

Q98. Solution

Correct Answer: (D)

Since, the system has no solution.

$$\begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & -1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow 2(-2\lambda + 1) + 1(\lambda + 1) + 2(3) = 0$$

$$\Rightarrow \quad -4\lambda + 2 + \lambda + 1 + 6 = 0$$

$$\Rightarrow \quad 3\lambda = 9 \Rightarrow \lambda = 3$$

Q99. Solution

Correct Answer: (D)

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore |A| = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= 1(1+3) + 1(2+3) + 1(2-1) = 4 + 5 + 1 = 10$$

$$\Rightarrow \text{Adj } A = \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}^T = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$\Rightarrow \quad B = A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix} \left[\because A^{-1} = \frac{1}{|A|} \text{Adj } A\right]$$

$$\Rightarrow 10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

Hence, $\alpha = 5$

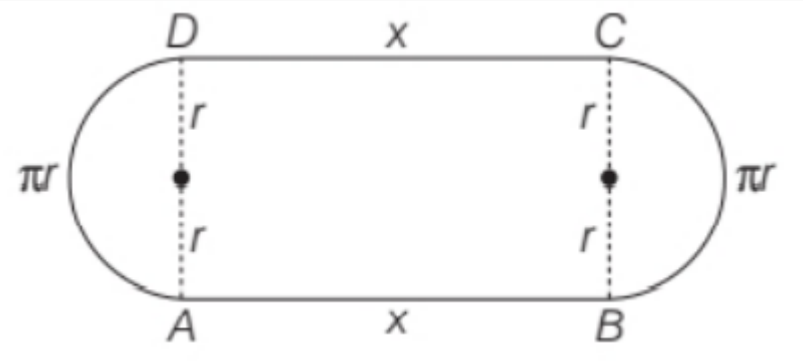
Q100. Solution

Correct Answer: (A)

$$y=\cos^{-1}\left(\frac{7}{2}(1+\cos 2x)+\sqrt{\left(\sin^2 x-48\cos^2 x\right)}\sin x\right)$$
$$=\cos^{-1}\left((7\cos x)(\cos x)+\sqrt{1-49\cos^2 x}\sqrt{1-\cos^2 x}\right)$$
$$=\cos^{-1}(\cos x)-\cos^{-1}(7\cos x)\quad[\because \cos x<7\cos x]$$
$$=x-\cos^{-1}(7\cos x)$$

Q101. Solution

Correct Answer: (C)



Perimeter $=440\text{ft}$

$\Rightarrow\quad 2x+\pi r+\pi r=440$

$2x+2\pi r=440$

$A=$ Area of the rectangular portion $=x2r$

$$A=x\frac{(440-2x)}{\pi}=\frac{1}{\pi}\left(440x-2x^2\right)$$

Let $\frac{dA}{dx}=\frac{1}{\pi}(440-4x)=0$

$\Rightarrow x=110$ for which $\dfrac{d^2A}{dx^2}<0$

$\Rightarrow A$ is maximum when $x=110$

$$\Rightarrow 2r=\frac{440-2x}{\pi}=\frac{440-220}{22/7}=70$$

$\therefore\quad r=35\text{ft}$ and $x=110\text{ft}$

Q102. Solution

Correct Answer: (A)

Given that

$$\left(\frac{dy}{dx}\right)\tan x = y\sec^2 x + \sin x$$

$$\tan x\left(\frac{dy}{dx}\right) - y\sec^2 x = \sin x$$

$$\frac{dy}{dx} + \left(\frac{-\sec^2 x}{\tan x}\right)y = \frac{\sin x}{\tan x}$$

$$\text{Here, } P = -\frac{\sec^2 x}{\tan x} \text{ and } Q = \cos x$$

$$\text{IF } = e^{\int P dx} = e^{\int \frac{-\sec^2 x}{\tan x} dx}$$

$$\text{IF } = e^{-\log(\tan x)} = e^{\log \cot x} = \cot x$$

$$\text{Now, } y \times \text{IF} = \int (Q \times \text{IF}) dx + C$$

$$y \cdot \cot x = \int \cos x \cdot \cot x + C$$

$$y \cdot \cot x = \int \frac{1 - \sin^2 x}{\sin x} + C$$

$$y \cdot \cot x = \log|\operatorname{cosec} x - \cot x| + \cos x + C$$

Q103. Solution

Correct Answer: (D)

Solving the given equations,

$$(mx + c)^2 = 4ax - 4a^3$$

$$\Rightarrow m^2x^2 + 2mc \cdot x + c^2 = 4ax - 4a^3$$

$$\Rightarrow m^2x^2 + (2mc - 4a)x + c^2 + 4a^3 = 0$$

Since the straight line touches the parabola at a point, so, the discriminant $= 0$

$$\Rightarrow \quad (2mc - 4a)^2 - 4m^2\left(c^2 + 4a^3\right) = 0$$

$$\Rightarrow 4m^2c^2 - 16amc + 16a^2 - 4m^2c^2 - 16a^3m^2 = 0$$

$$\Rightarrow -mc + a - a^2m^2 = 0$$

$$\Rightarrow mc = a - a^2m^2 \Rightarrow c = \frac{a}{m} - a^2m$$

Q104. Solution

Correct Answer: (C)

For the parabola $y^2 = 4ax$, the equation of normal at $P\left(am^2, -2am\right)$ is $y = mx - 2am - am^3$.

Here, $m = \tan 60^\circ = \sqrt{3}$

$$\therefore P \equiv \left(a(\sqrt{3})^2, -2a(\sqrt{3})\right) \equiv (6, -4\sqrt{3}) \quad (\because a = 2)$$

Thus, P satisfies $2x - y - 12 - 4\sqrt{3} = 0$

Q105. Solution

Correct Answer: (C)

$I=\int \frac{1}{\left(\frac{x-1}{x+2}\right)^{3/4}(x+2)^2}dx$

Let $\frac{x-1}{x+2}=t \Rightarrow \frac{3dx}{(x+2)^2}=dt$

$$\Rightarrow \quad I=\frac{1}{3}\int \frac{1}{t^{3/4}}dt=\frac{1}{3}\left(\frac{t^{1/4}}{1/4}\right)+C$$

$$=\frac{4}{3}t^{1/4}+C=\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$$

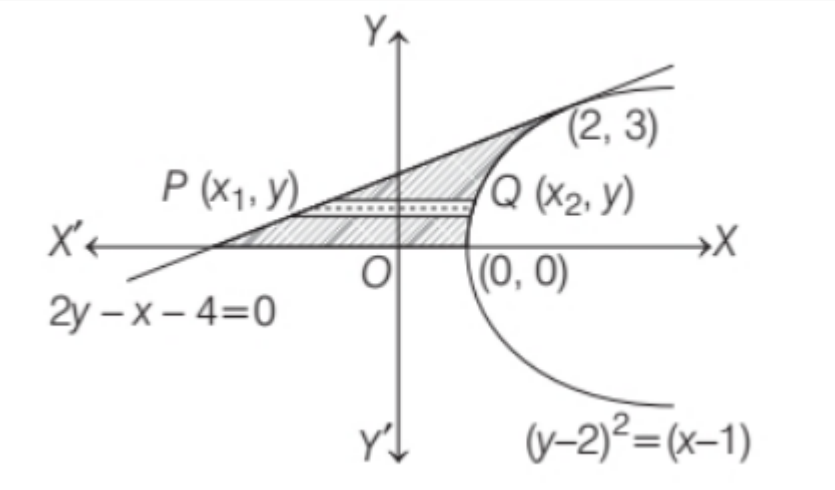
Q106. Solution

Correct Answer: (C)

The equation of the parabola is $y^2-4y-x+5=0$ The equation of tangent at $(2,3)$ is

$$3y-2(y+3)-\frac{x+2}{2}+5=0$$

$$2y-x-4=0$$



$\therefore$ Required area A is given by

$$A=\int_0^3 (x_2-x_1)dy$$

$$\Rightarrow \quad A=\int_0^3 \left[\left\{(y-2)^2+1\right\}-\{2y-4\}\right]dy$$

$$\Rightarrow \quad A=\int_0^3 \left(y^2-6y+9\right)dy=\int_0^3 (3-y)^2dy$$

$$=-\left[\frac{(3-y)^3}{3}\right]_0^3=9$$

Q107. Solution

Correct Answer: (D)

Given that, $\frac{\hat{\mathbf{u}}+\hat{\mathbf{v}}}{2}+\hat{\mathbf{u}} \times \hat{\mathbf{v}} \ =1$

$\Rightarrow \quad \frac{\hat{\mathbf{u}}+\hat{\mathbf{v}}}{2}+\hat{\mathbf{u}} \times \hat{\mathbf{v}}^{\ 2}=1$

$\Rightarrow \quad \frac{2+2 \cos \theta}{4}+\sin^2 \theta=1$

$[\because \hat{\mathbf{u}} \cdot(\hat{\mathbf{u}} \times \hat{\mathbf{v}})=\hat{\mathbf{v}}(\hat{\mathbf{u}} \times \hat{\mathbf{v}})=0]$

$\Rightarrow \quad \cos^2 \frac{\theta}{2}=\cos 2\theta$

$\Rightarrow \quad \theta=n\pi \pm \frac{\theta}{2}, n \in z$

$\Rightarrow \quad \theta=\frac{2\pi}{3}$

$\Rightarrow \quad |\hat{\mathbf{u}} \times \hat{\mathbf{v}}|=\sin \frac{2\pi}{3}=\sin \frac{\pi}{3}=\ \frac{\hat{\mathbf{u}}-\hat{\mathbf{v}}}{2}$

Q108. Solution

Correct Answer: (B)

Let P be the probability of getting an even number. Then by hypothesis the probability of getting an odd number is $3P$. Since the events of getting an even number and an odd number are mutually exclusive and exhaustive.

$\therefore \quad P+3P=1 \Rightarrow P=1/4$

Thus, the probability of getting an odd number in a single throw is $\frac{3}{4}$ and that of an even number is $\frac{1}{4}$.

If the die is thrown twice, then the sum of the numbers in two throws is even if both the numbers are even or both are odd.

$\therefore$ Required probability $=\frac{3}{4} \times \frac{3}{4}+\frac{1}{4} \times \frac{1}{4}=\frac{10}{16}=\frac{5}{8}$

Q109. Solution

Correct Answer: (B)

$$2 \cos \theta\left[\cos 120^{\circ}+\cos 2\theta\right]=1$$

$$\Rightarrow 2 \cos \theta\left(-\frac{1}{2}+2 \cos^2 \theta-1\right)=1$$

$$\Rightarrow 4 \cos^3 \theta-3 \cos \theta-1=0$$

$$\Rightarrow \cos 3\theta=1=\cos 0$$

$$\Rightarrow 3\theta=2n\pi \text{ or } \theta=\frac{2n\pi}{3}, n \in z$$

Given the values so that $2n$ does not exceed 18 .

Hence, the sum $=\frac{2\pi}{3} \sum_1^9 n=\frac{2\pi}{3} \times \frac{9(9+1)}{2}=30\pi$

Q110. Solution

Correct Answer: (C)

We have,

$16^{\sin^2\theta} + 16^{\cos^2\theta} = 10$

$\Rightarrow \quad 16^{\sin^2\theta} + 16^{1-\sin^2\theta} = 10$

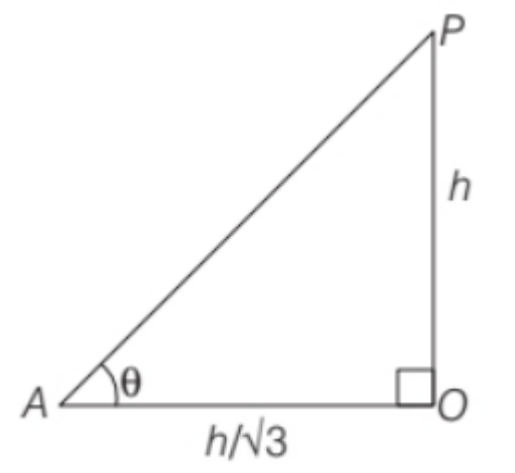
$\Rightarrow x + \frac{16}{x} = 10$, where $x = 16^{\sin^2\theta}$

$\Rightarrow \quad x^2 - 10x + 16 = 0 \Rightarrow x = 2, 8$

$\therefore \quad 16^{\sin^2\theta} = 2, 8 \Rightarrow 2^{4\sin^2\theta} = 2, 2^3$

$\Rightarrow \quad 4\sin^2\theta = 1, 3 \Rightarrow \sin^2\theta = \frac{1}{4}, \left(\frac{\sqrt{3}}{2}\right)^2$

$\Rightarrow \quad \sin\theta = \frac{1}{2}, \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{\pi}{3}$



Let the altitude of the sun be θ. Then,

$$\tan\theta = \frac{h}{h/\sqrt{3}} \Rightarrow \tan\theta = \sqrt{3}$$

$$\Rightarrow \quad \theta = \pi/3 \Rightarrow \theta = 2\alpha$$

Q111. Solution

Correct Answer: (B)

Suppose the parabola $y = x^2 + px + q$ cuts X-axis at $A(\alpha, 0)$ and $B(\beta, 0)$.
Then, α, β are roots of the equation $x^2 + px + q = 0$
$\therefore \quad \alpha + \beta = -p$ and $\alpha\beta = q$
The parabola $y = x^2 + px + q$ cuts Y-axis at $(0, q)$.
Let the equation of the circle passing through A, B and C be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \qquad \ldots(i)$$
$$\therefore \alpha^2 + 2g\alpha + c = 0 \qquad \ldots(ii)$$
$$\beta^2 + 2g\beta + c = 0 \qquad \ldots(iii)$$
$$\text{and } q^2 + 2fq + c = 0 \qquad \ldots(iv)$$

Subtracting Eq. (iii) from Eq. (ii), we get

$$\alpha + \beta + 2g = 0$$
$$\Rightarrow \quad g = p/2$$

Adding Eqs. (ii) and (iii), we get

$$\alpha^2 + \beta^2 + 2g(\alpha + \beta) + 2c = 0$$
$$(\alpha + \beta)^2 - 2\alpha\beta + 2g(\alpha + \beta) + 2c = 0$$
$$p^2 - 2q - p^2 + 2c = 0$$

$\left[\because \alpha + \beta = p \text{ and } g = \frac{p}{2}\right]$

Putting $c = q$ in Eq. (iv), we get $f = -\left(\frac{q+1}{2}\right)$

Substituting the values of g, f and c in Eq. (i), we obtain the equation of family of circles passing through A, B and C as

$x^2 + y^2 + px - (q + 1)y + q = 0$

Clearly, it passes through $(0, 1)$.

Q112. Solution

Correct Answer: (C)

We can arrange the letters $\mathrm{H, A, A, A}$ in $\frac{4!}{3!} = 4$ ways. If one possible arrangement is

XXXX

Then, we can arrange $\mathrm{V, N}$ at any of the two places marked with O in the following arrangement

$\mathrm{O} \times \mathrm{O} \times \mathrm{O} \times \mathrm{O} \times \mathrm{O}$

Thus, we can arrange V and N in ${}^5P_2 = 20$ ways. Thus, the number of ways in which letters can be arranged is $4 \times 20 = 80$

Q113. Solution

Correct Answer: (D)

It is given that $a_1, a_2, a_3 \ldots$ are in HP. Therefore, $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \ldots\ldots\ldots$ are in AP. Let d be the common difference of the AP.

$\therefore \quad \frac{1}{a_n} = \frac{1}{a_1} + (n-1)d$ and $\frac{1}{a_{20}} = \frac{1}{a_1} + 19d$

$\Rightarrow \quad \frac{1}{a_n} = \frac{1}{5} + (n-1)d$ and $\frac{1}{25} = \frac{1}{5} + 19d$

$\Rightarrow \quad \frac{1}{a_n} = \frac{1}{5} + (n-1)d$ and $d = \frac{-4}{19 \times 25}$

$\Rightarrow \quad \frac{1}{a_n} = \frac{1}{5} - \frac{4(n-1)}{19 \times 25}$

$\Rightarrow \quad \frac{1}{a_n} = \frac{95-4n+4}{19 \times 25}$

$\Rightarrow \quad a_n = \frac{99-4n}{19 \times 25}$

Now, $a_n < 0$

$\Rightarrow \quad \frac{99-4n}{19 \times 25} < 0$

$\Rightarrow \quad 99 - 4n < 0$

$\Rightarrow \quad 4n > 99$

$\Rightarrow \quad n > 24\frac{3}{4} \Rightarrow n \geq 25$

Q114. Solution

Correct Answer: (B)

Given plane $3x + y + 2z + 6 = 0$

and line $\frac{x-\frac{1}{3}}{\frac{2b}{3}} = \frac{y-3}{-1} = \frac{z-1}{a}$

Since, plane is parallel to line, then

$$3\left(\frac{2b}{3}\right) + (1)(-1) + 2(a) = 0$$

$2b - 1 + 2a = 0$

$$\Rightarrow \quad a + b = \frac{1}{2}$$

Now, $3a + 3b = \frac{3}{2}$

Q115. Solution

Correct Answer: (D)

Since, $a+b=-p, ab=1$ .....(i)

and $c+d=-q, cd=1$

Now $(a-c)(b-c)$ and $(a+d)(b+d)$ are the roots of $x^2+ax+\beta=0$

$(a-c)(b-c)(a+d)(b+d)=\beta$

$\Rightarrow \left(ab-ac-bc+c^2\right)\left(ab+ad+bd+d^2\right)=\beta$

$\Rightarrow \left\{1-c(a+b)+c^2\right\}\left\{1+d(a+b)+d^2\right\}=\beta$

$\Rightarrow \quad \left(1+pc+c^2\right)\left(1-pd+d^2\right)=\beta$

$\Rightarrow 1-pd+d^2+pc-p^2cd+pcd^2+c^2-pc^2d$

$+c^2d^2=\beta$

$\Rightarrow 1-pd+d^2+pc-p^2+pd+c^2-pc+1=\beta$

$[\because cd=1]$

$\Rightarrow \quad 2+d^2+c^2-p^2=\beta$

$$\Rightarrow \quad 2cd+c^2+d^2-p^2=\beta$$
$$(c+d)^2-p^2=\beta$$
$$q^2-p^2=\beta \quad [\because (c+d)=-q]$$

Q116. Solution

Correct Answer: (C)

Given,

$$\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}\begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}^{-1}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}\cdot\frac{1}{1+\tan^2\theta}\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}$$

$$=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \frac{1}{1+\tan^2\theta}\begin{bmatrix} 1-\tan^2\theta & -2\tan\theta \\ 2\tan\theta & 1-\tan^2\theta \end{bmatrix}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} \frac{1-\tan^2\theta}{1+\tan^2\theta} & \frac{2\tan\theta}{1+\tan^2\theta} \\ \frac{2\tan\theta}{1+\tan^2\theta} & \frac{1-\tan^2\theta}{1+\tan^2\theta} \end{bmatrix}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \quad a=\cos 2\theta, b=\sin 2\theta$$

Q117. Solution

Correct Answer: (A)

Let $y=(1+x)^{1/x}$

$\Rightarrow \log y=\frac{1}{x}\log(1+x)=\frac{1}{x}\left[x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ldots\right]$

$=1-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\ldots$

$y=e^{\left(1-\frac{x}{2}+\frac{x^2}{3}+\ldots\right)}=ee^{\left(\frac{-x}{2}+\frac{x^2}{3}\ldots\ldots\right)}$

$$=e\left[1+\left(-\frac{x}{2}+\frac{x^2}{3}-\ldots\right)+\frac{1}{2!}\left(-\frac{x}{2}+\frac{x^2}{3}-\ldots\right)^2+\ldots\right]$$

$y-e+\frac{1}{2}ex=ex^2\left[\frac{1}{3}+0(x)+\frac{1}{2}\left(-\frac{1}{2}+0(x)\right)^2+\ldots.\right]$

$[\because 0(x)$ in terms containing $x]$

$\lim \frac{y-e+\frac{1}{2}ex}{x^2}=e\left[\frac{1}{3}+\frac{1}{8}\right]=\frac{11}{24}e$

Q118. Solution

Correct Answer: (A)

Let $P\left(t, \frac{4t-20}{5}\right)$ be a point on the line

$4x-5y=20$. Then the chord of contact of tangents drawn from P to the circle $x^2+y^2=9$ is

$tx+\left(\frac{4t-20}{5}\right)y=9 \qquad \ldots(i)$

Let $\theta(h,k)$ be the mid-point of this chord of contact, then its equation is also $hx+ky=h^2+k^2 \qquad \ldots(ii)$

[using $T=S'$]

Clearly Eqs. (i) and (ii) represent the same line

$$\therefore \frac{t}{h}=\frac{4t-20}{5k}=\frac{9}{h^2+k^2}$$

$$\Rightarrow \frac{t}{h}=\frac{9}{h^2+k^2} \text{ and } \frac{t}{h}=\frac{4t-20}{5k}$$

$$\Rightarrow t=\frac{9h}{h^2+k^2} \text{ and } t=\frac{20h}{4h-5k}$$

$$\Rightarrow \frac{9h}{h^2+k^2}=\frac{20h}{4h-5k}$$

$$\Rightarrow h\left\{20\left(h^2+k^2\right)-36h+45k\right\}=0$$

$x=0,$ or $\left[20\left(x^2+y^2\right)-36x+45y\right]=0$

Q119. Solution

Correct Answer: (B)

$f(x)=\int \frac{x^2 dx}{(1+x^2)\left(1+\sqrt{1+x^2}\right)}$

Let $x=\tan\theta$

$\Rightarrow \quad dx=\sec^2\theta d\theta=\left(1+x^2\right)d\theta$

$f(x)=\int \frac{x^2 dx}{(1+x^2)\left(1+\sqrt{1+x^2}\right)}$

$=\int \frac{\tan^2\theta\sec^2\theta d\theta}{\sec^2\theta(1+\sec\theta)}=\int \frac{\tan^2\theta d\theta}{1+\sec\theta}$

$=\int \frac{\sin^2\theta d\theta}{\cos\theta(1+\cos\theta)}$

$=\int \frac{1-\cos^2\theta d\theta}{\cos\theta(1+\cos\theta)}$

$=\int \frac{(1-\cos\theta)d\theta}{\cos\theta}$

$=\int \sec\theta d\theta-\int d\theta$

$=\log\left(x+\sqrt{1+x^2}\right)-\tan^{-1}x+C$

$\therefore \quad f(0)=\log(0+\sqrt{1+0})-\tan^{-1}(0)+C$

$0=\log 1-0+C$

$\Rightarrow \quad C=0$

$\therefore \quad f(1)=\log\left(1+\sqrt{1+1^2}\right)-\tan^{-1}(1)$

$=\log(1+\sqrt{2})-\frac{\pi}{4}$

Q120. Solution

Correct Answer: (C)

Let the two unknown items be x and y. Then,

Mean $=4 \Rightarrow \frac{1+2+6+x+y}{5}=4$

$\Rightarrow \quad x+y=11\ldots(i)$

and variance $=5.2$

$\Rightarrow \frac{1^2+2^2+6^2+x^2+y^2}{5}-(\text{ mean })^2=5.2$

$\Rightarrow \quad 41+x^2+y^2=5(5.2+16)$

$\Rightarrow \quad 41+x^2+y^2=106$

$\Rightarrow \quad x^2+y^2=65$

On solving Eqs. (i) and (ii), we get

$x=4, y=7$ or $x=7, y=4$

Q121. Solution

Correct Answer: (A)

Since, the first 11 terms are in AP, $d = 2$

$\therefore \quad a_{11} = a + 10d = a + 20$

The middle term of AP is

$T_6 = a + 5d = a + 10$

For the next 11 terms in GP

$r = 2$

$\therefore$ The middle term of GP is $b(2)^5$ where, b is the first term of a GP which is the last term of AP

$b(2)^5 = (a + 20)32$

According to the given condition,

$$\Rightarrow \quad a + 10 = (a + 20)32$$

$$\Rightarrow \quad 31a = 10 - 640$$

$$a = -\frac{630}{31}$$

$\therefore$ Middle term of entire sequence is $11th$ term

$$\therefore \quad T_{11} = \frac{-630}{31} + 10 \times d$$

$$= \frac{-630}{31} + 10 \times 2 = \frac{-10}{31}$$

Q122. Solution

Correct Answer: (B)

As the given system of equations has a non-trivial solution.

$$\Delta = \begin{vmatrix} p & a & a \\ b & q & b \\ c & c & r \end{vmatrix} = 0$$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$

$$\Delta = \begin{vmatrix} p & a-p & a-p \\ b & q-b & 0 \\ c & 0 & r-c \end{vmatrix} = 0$$

Expanding along C_3, we get

$$(a-p)\begin{vmatrix} b & q-b \\ c & 0 \end{vmatrix} + (r-c)\begin{vmatrix} p & a-p \\ b & q-b \end{vmatrix} = 0$$

$$\Rightarrow (a-p)(-c)(q-b) + (r-c)$$

$$\{p(q-b) - b(a-p)\} = 0$$

$$\Rightarrow$$

$$(p-a)(q-b)c + p(r-c)(q-b) + b(r-c)(p-a) = 0$$

Dividing by $(p-a)(q-b)(r-c)$, we get

$$\frac{c}{r-c} + \frac{p}{p-a} + \frac{b}{q-b} = 0$$

$$\Rightarrow \frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = \frac{q-b}{q-b} + \frac{r-c}{r-c} = 2$$

Q123. Solution

Correct Answer: (A)

We have, $\frac{1}{2} gof(x) = 2x^2 - 5x + 2$

$\Rightarrow g(f(x)) = 4x^2 - 10x + 4$

$\Rightarrow (f(x))^2 + f(x) - 2 = 4x^2 - 10x + 4$

$\Rightarrow (f(x))^2 + f(x) - \left(4x^2 - 10x + 6\right) = 0$

$$\Rightarrow f(x) = \frac{-1 \pm \sqrt{1 + 4\left(4x^2 - 10x + 6\right)}}{2}$$

$$\Rightarrow f(x) = \frac{-1 \pm \sqrt{16x^2 - 40x + 25}}{2}$$

$$\Rightarrow f(x) = \frac{-1 \pm (4x - 5)}{2} = 2x - 3, -2x + 2$$

Hence, $f(x) = 2x - 3$

Q124. Solution

Correct Answer: (A)

$\left[\frac{1+\sin\frac{\pi}{8}+i\cos\frac{\pi}{8}}{1+\sin\frac{\pi}{8}-i\cos\frac{\pi}{8}}\right]^n = \left[\frac{1+\cos\alpha+i\sin\alpha}{1+\cos\alpha-i\sin\alpha}\right]^n$

$\left[\because \alpha = \frac{\pi}{2} - \frac{\pi}{8}\right]$

$= \left[\frac{2\cos^2\frac{\alpha}{2}+2i\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}{2\cos^2\frac{\alpha}{2}-2i\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}\right]^n$

$= \left[\frac{\cos\frac{\alpha}{2}+i\sin\frac{\alpha}{2}}{\cos\frac{\alpha}{2}-i\sin\frac{\alpha}{2}}\right]^n = \left(e^{2i\frac{\alpha}{2}}\right]^n = e^{in\alpha}$

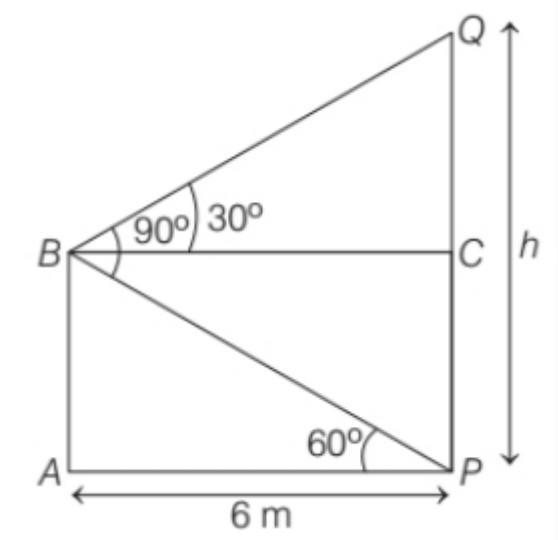
$= e^{in\left(\frac{3\pi}{8}\right)} = \cos\frac{3n\pi}{8} + i\sin\frac{3n\pi}{8}$

For $n = 4$ we get imaginary part.

Q125. Solution

Correct Answer: (A)

Let PQ be the house subtending a right angle at the window B of opposite house AB



In $\triangle PAB$, we have

$\tan 60^\circ = \frac{AB}{6} = AB = 6\sqrt{3}$ m

In $\triangle CBQ$, we have

$\tan 30^\circ = \frac{h-CP}{BC}$

$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{h-6\sqrt{3}}{6} \quad [\because AB = CP, BC = AP]$

$\Rightarrow \quad h = 6\left(\sqrt{3} + \frac{1}{\sqrt{3}}\right)$

$\Rightarrow \quad h = 8\sqrt{3}$ m

Q126. Solution

Correct Answer: (C)

$V = \frac{4}{3}\pi r^3 \Rightarrow 4500\pi = \frac{4\pi r^3}{3}$

$\Rightarrow \quad r = 15$ m

After 49 min $= (4500 - 49.72)\pi$

$= 972\pi \mathrm{m}^3$

$$\Rightarrow \quad 972\pi = \frac{4}{3}\pi r^3$$

$$\Rightarrow \quad r^3 = 3 \times 243 = 3 \times 3^5$$

$r = 9$

$$\frac{dv}{dt} = 4\pi r^2 \left(\frac{dr}{dt}\right)$$

$$72\pi = 4\pi \times 9 \times 9 \left(\frac{dr}{dt}\right)$$

$$\frac{dr}{dt} = \left(\frac{2}{9}\right)$$

Q127. Solution

Correct Answer: (D)

$\frac{\sin 3B}{\sin B} = \frac{3\sin B - 4\sin^3 B}{\sin B}$

$= 3 - 4\sin^2 B$

$= 3 - 4\left(1 - \cos^2 B\right)$

$= -1 + \frac{4\left(a^2+c^2-b^2\right)^2}{4(ac)^2}$

$= -1 + \frac{\left(\frac{a^2+c^2}{2}\right)^2}{(ac)^2}$

$= -1 + \frac{\left(a^2+c^2\right)^2}{4(ac)^2}$

$= \left(\frac{c^2-a^2}{2ac}\right)^2$

Q128. Solution

Correct Answer: (B)

Given, sum of the coefficient $= 1024$

i.e. $\quad 2^n = 1024 = 2^{10}$

$\Rightarrow \quad n = 10$

Since, n is even, so greatest coefficient

$= {}^nC_{n/2} = {}^{10}C_5 = 252$

Q129. Solution

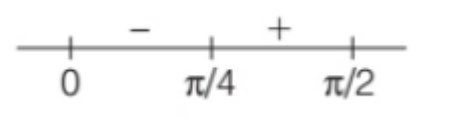
Correct Answer: (B)

Given, $y = \sin x + \cos x, x \in [0, \pi/2]$

$\frac{dy}{dx} = \cos x - \sin x$

$y = |\cos x - \sin x|$

$$= \begin{bmatrix} \cos x - \sin x & x \in [0, \pi/4] \\ \sin x - \cos x & x \in [\pi/4, \pi/2] \end{bmatrix}$$

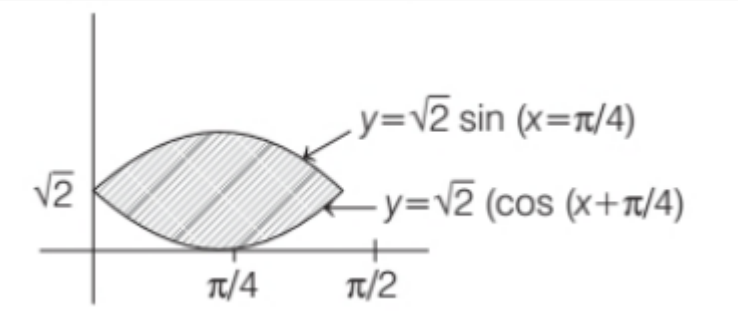


$\therefore$ Required area

$$= \int_0^{\pi/4} |(\sin x + \cos x) - (\cos x - \sin x)| dx + \int_{\pi/4}^{\pi/2} |2 \cos x| dx$$

$= \int_0^{\pi/4} |2 \sin x| dx + \int_{\pi/4}^{\pi/2} |2 \cos x| dx$

$= 2(-\cos x)_0^{\pi/4} + 2(\sin x)_{\pi/4}^{\pi/2}$



$= 2\left[-\frac{1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}}\right]$

$$=2\left(2 - \frac{2}{\sqrt{2}}\right)$$

$=2(2 - \sqrt{2})$

$=4 - 2\sqrt{2}$

$=2\sqrt{2}(\sqrt{2} - 1)$

Q130. Solution

Correct Answer: (C)

$\frac{\sin\alpha - \sin\gamma}{\cos\gamma - \cos\alpha} = \frac{2\cos\frac{\alpha+\gamma}{2}\sin\frac{\alpha-\gamma}{2}}{2\sin\frac{\alpha+\gamma}{2}\cdot\sin\frac{\alpha-\gamma}{2}}$

$= \cot\frac{\alpha+\gamma}{2}$

$$\begin{bmatrix} \because \sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right) \\ \text{and } \cos A - \cos B = 2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{B-A}{2}\right) \end{bmatrix}$$

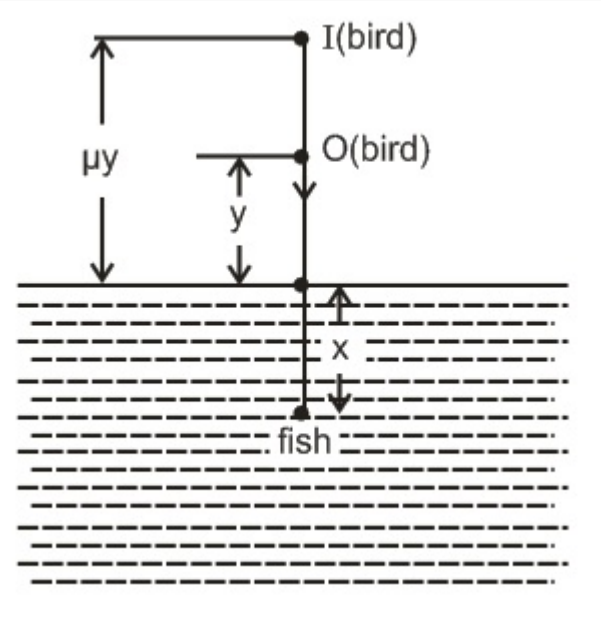
But α, β, γ are in AP

$\Rightarrow \quad \frac{\alpha+\gamma}{2} = \beta$

So, required value is $\cot\beta$.

Q131. Solution

Correct Answer: (A)



distance of bird appeared to fish

$\mathrm{h} = \mathrm{x} + \mu_{\mathrm{y}}$

Differentiating w.r.t. to time $\Rightarrow \frac{\mathrm{dh}}{\mathrm{dt}} = \frac{\mathrm{dx}}{\mathrm{dt}} + \mu \frac{\mathrm{dy}}{\mathrm{dt}} \Rightarrow \frac{\mathrm{dh}}{\mathrm{dt}} = \frac{4}{3} \times 6$

$\frac{\mathrm{dh}}{\mathrm{dt}} = 8 \ \mathrm{m/s}$

Q132. Solution

Correct Answer: (B)

Pressure of gas is given by, $P = \frac{1}{3}\left(\frac{m}{V}\right)c^2$, where $m,\ V$ and c are mass, volume and R.M.S. speed of gas molecule.

Keeping the volume same, ratio of two states is given by formula,

$\Rightarrow \frac{P_1}{P_2} = \frac{\frac{1}{3}\left(\frac{m_1}{V}\right){c_1}^2}{\frac{1}{3}\left(\frac{m_2}{V}\right){c_2}^2}$

$\Rightarrow \frac{P_1}{P_2} = \frac{m_1{c_1}^2}{m_2{c_2}^2}$

Substituting the given data, we have

$\Rightarrow \frac{P_1}{P_2} = \frac{m_1{c_1}^2}{\left(\frac{m_1}{2}\right)(2c_1)^2}$

$\Rightarrow \frac{P_1}{P_2} = \frac{1}{2}$

Q133. Solution

Correct Answer: (A)

Energy of photoelectron is given by Einstein is, $KE = E - \phi$, where $E \& \phi$ represent energy of incident photon and work function of metal respectively.

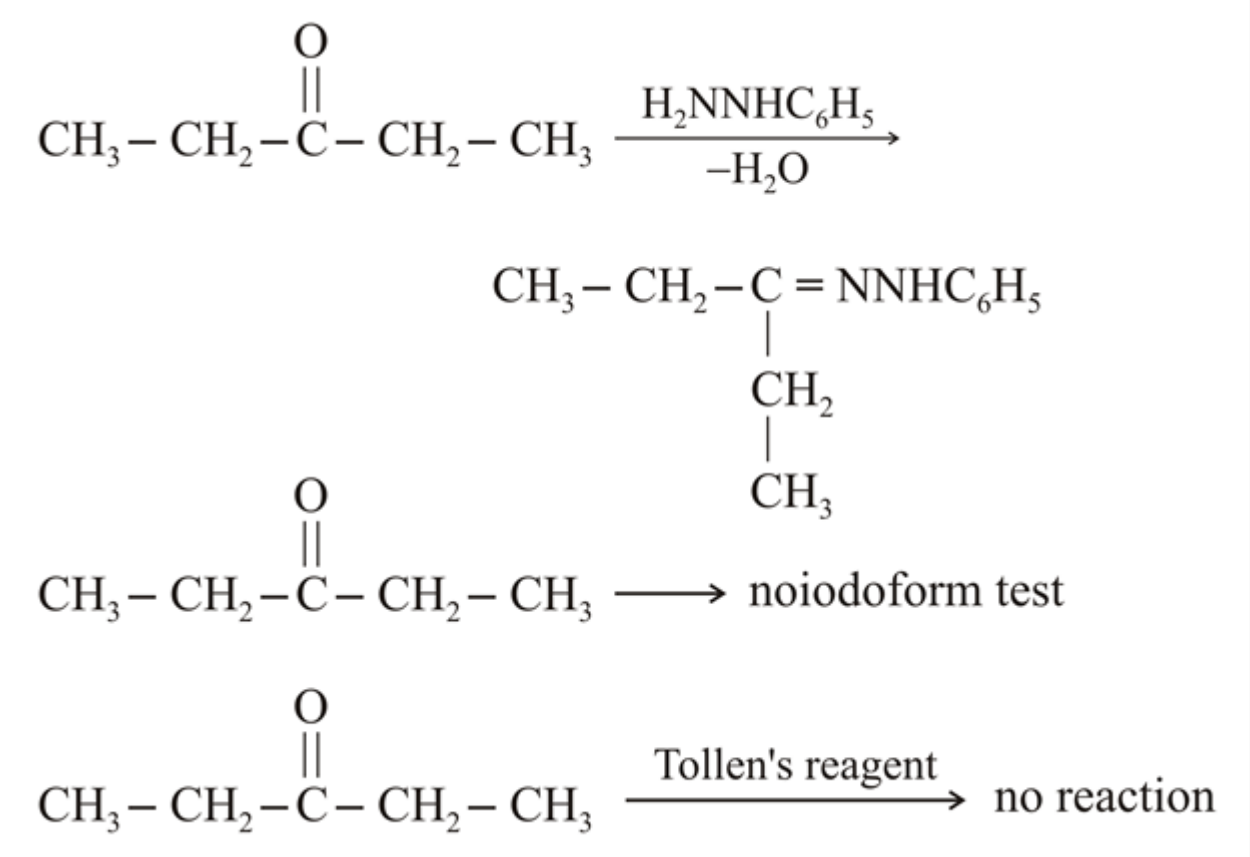
So kinetic energy is energy released the difference between energy of photon and work function of metal.

$\Rightarrow KE = E - \phi$

Q134. Solution

Correct Answer: (C)

The compound is pentanone-3



Q135. Solution

Correct Answer: (C)

For the reaction $2A + B_2 \rightleftharpoons 2AB$,

Rate $\propto [A]^x[B_2]^y$. On substituting the given data, we get

From experimental 1, $1.6 \times 10^{-4} \propto [0.50]^x[0.50]^y$ ...(i)

From experimental 2, $3.2 \times 10^{-4} \propto [0.50]^x[1.00]^y$ ...(ii)

From experimental 3, $3.2 \times 10^{-4} \propto [1.00]^x[1.00]^y$ ...(iii)

$1 = \left[\frac{1.00}{0.50}\right]^x$

or $1 = 2^x$ or $2^0 = 2^x$ or $x = 0$

Now, divide equation (ii) by equation (i) we get,

$2 = \left[\frac{1.00}{0.50}\right]^y$

$2 = 2^y$

$y = 1$

Thus rate equation is: Rate $= k[A]^0[B_2]^1 = k[B_2]$.

Q136. Solution

Correct Answer: (B)

$$P_4 + 3O_2 \xrightarrow{N_2} P_4O_6$$

N_2 is used to control further oxidation in the process.

Presence of nitrogen gas with oxygen gas will decrease the reactivity of oxygen gas and reaction is taking place with optimum rate.

Reaction with pure oxygen will be explosive so nitrogen control the reaction.

Q137. Solution

Correct Answer: (B)

In the figure given in the question, there is a relation between figures placed horizontally beside each other.

In the first row, the part which is present in the first figure is absent in the third figure.

Similarly, in the second row, the part in the first figure is again absent in the third figure.

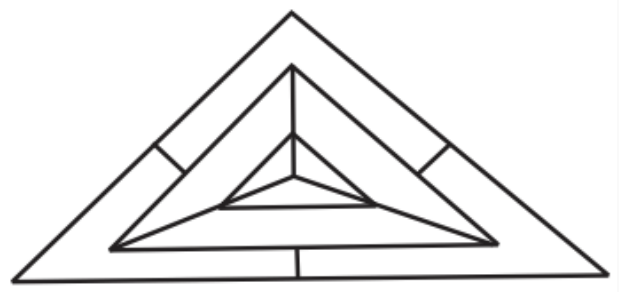
In the same way, third row figures will also follow the pattern and part in the first figure will be absent in the third figure. So, the third figure will contain only two diagonal lines as a cross.

Thus, option 2 is correct.

Q138. Solution

Correct Answer: (B)

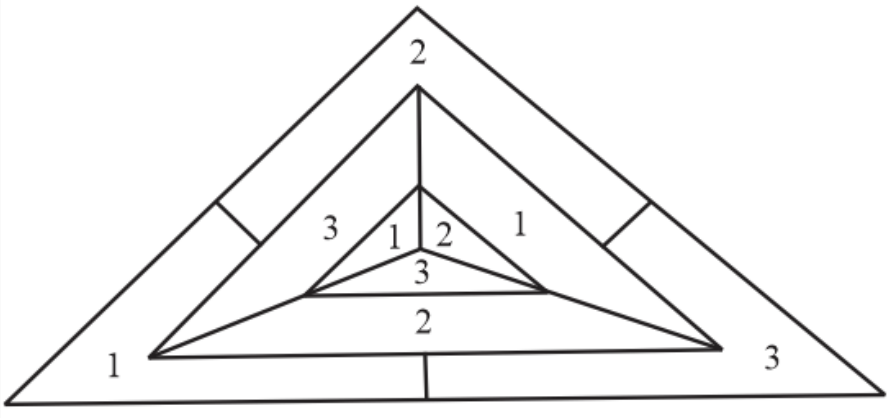
Given figure,



Here, we have to find out the minimum number of colors required to color the entire figure if no adjacent faces have the same color.

Here, we can see that each section to the given triangle is divided into three parts.

Now, if we pick three colors $1,\ 2$, and 3 to fill the triangle the following figure will be obtained,



Hence, the correct answer is 3.

Q139. Solution

Correct Answer: (A)

In the given series, we observe the pattern is the difference between consecutive numbers.

$20 - 11 = 938 - 20 = 1874 - 38 = 36$

So, the next difference is 72.

Thus, $74 + 72 = 146$ is the next number.

Hence, the correct answer is option A.

Q140. Solution

Correct Answer: (B)

Since, $A = \{x : -1 \leq x \leq 1\}$, $B = \{y : 1 \leq y \leq 2\}$ and

$y = f(x) = 1 + x^2$
For $x = -1,\ y = 1 + (-1)^2 = 2$
And for $x = 1,\ y = 1 + 1^2 = 2$
$\therefore f$ is not injective. (one-one)
Here,$\forall y \in B$, there is a preimage.
Hence, f is surjective.

Q141. Solution

Correct Answer: (B)

We have,

$I_n = \int_0^{\frac{\pi}{2}} x^n \sin x dx$

$\Rightarrow I_n = [-x^n \cos x]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} nx^{n-1} \cos x dx$

$\Rightarrow I_n = n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x dx$

$\Rightarrow I_n = n\left[\left[x^{n-1} \sin x\right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (n-1)x^{n-2} \sin x dx\right]$

$\Rightarrow I_n = n\left[\left(\frac{\pi}{2}\right)^{n-1} - (n-1)I_{n-2}\right]$

Hence,

$I_n + n(n-1)I_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1}$

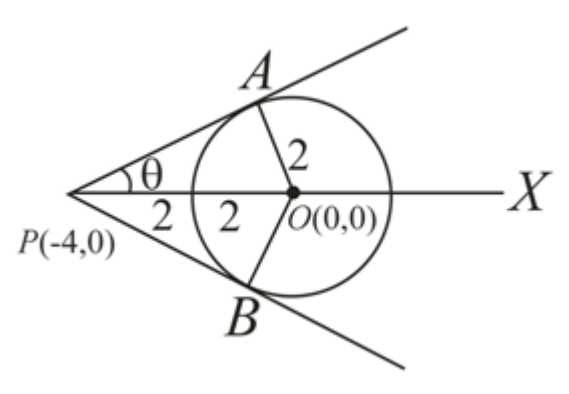
Q142. Solution

Correct Answer: (C)

From the Figure, $\sin\ \theta = \frac{OA}{OP} = \frac{2}{4} = \frac{1}{2}$

So,ar $(\Delta \mathrm{POA}) = \frac{1}{2} \cdot 2 \cdot 4 \cdot \sin\ 60^{\circ}$

$\therefore\ ar\ (\text{quad. } PAOB)$

$= 2 \cdot \frac{1}{2} \cdot 2 \cdot 4 \cdot \sin\ 60^{\circ}$

$= 8 \cdot \frac{\sqrt{3}}{2} = 4\sqrt{3}.$